



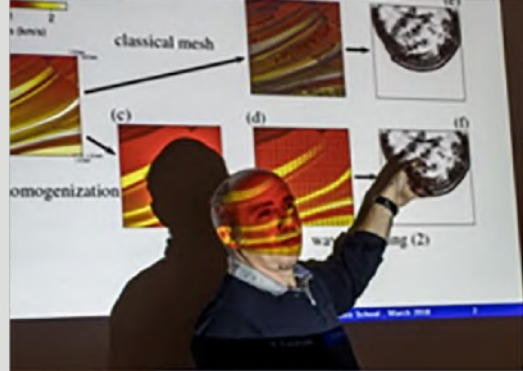
16th Munich Earth Science School
ObsPy on the Rocks - Cryoseismology

Welcome

to Sudelfeld

Heiner Igel & Thomas Lecocq

22 February 2026







16th Munich Earth Science School
ObsPy on the Rocks - Cryoseismology

Welcome

You are here



Try Pitch

Name Password

... a bit of history ... skience since 2011

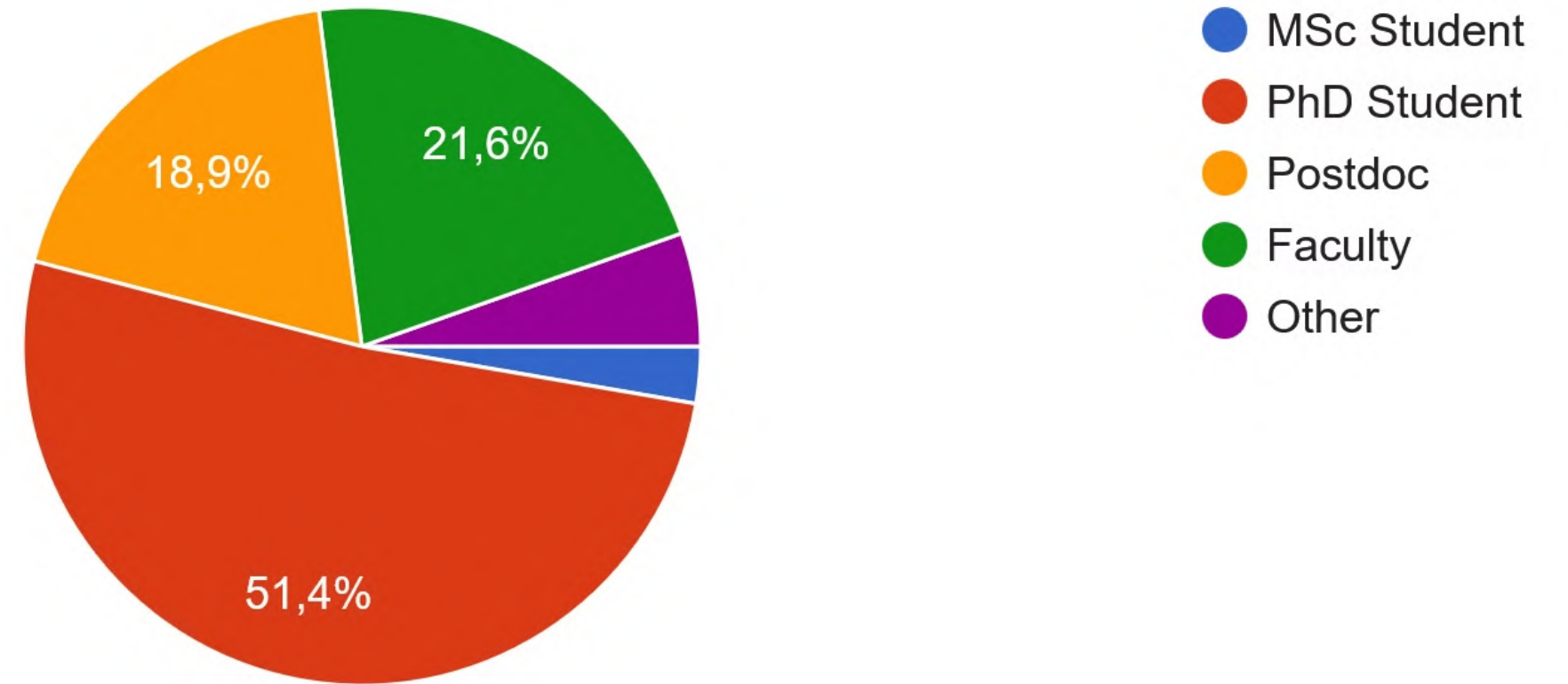
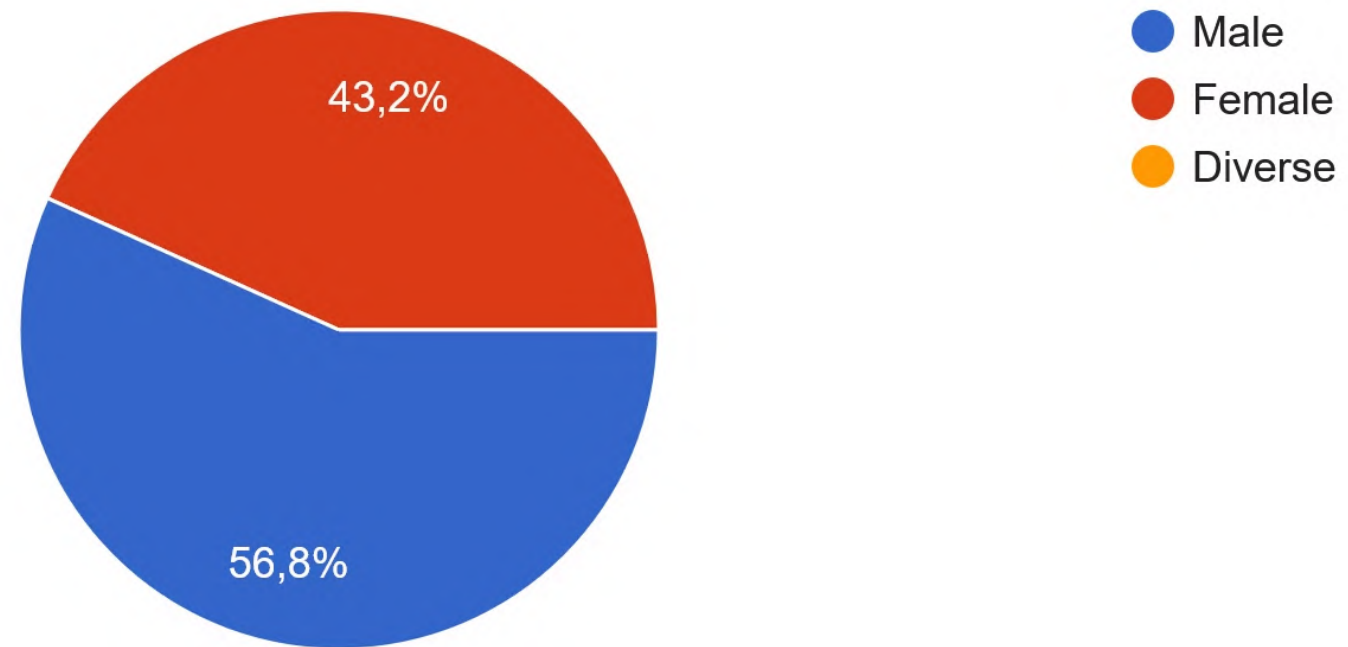
2011 - 1	ObsPy, Forward modelling, DG, SEM, Adjoint
2012 - 2	ObsPy, Source Inversion
2013 - 3	Python, ObsPy, Databases
2014 - 4	ObsPy, Seismic Array Processing
2015 - 5	ObsPy, Forward Modelling, FD,PS,FE,SE,FV,DG
2016 - 6	ObsPy, Induced Seismicity
2017 - 7	Simulation, Spectral Elements
2018 - 8	Wavefield gradients
2019 - 9	Machine Learning in Earth Sciences
2020 - 10	Inverse problems
2021 & 2022 - 11 & 12	
2023 - 13 (11)	Environmental Seismology
2024 - 14 (12)	Volcano Seismology
2025 - 15 (13)	Seismology in Cities

Hotel Information

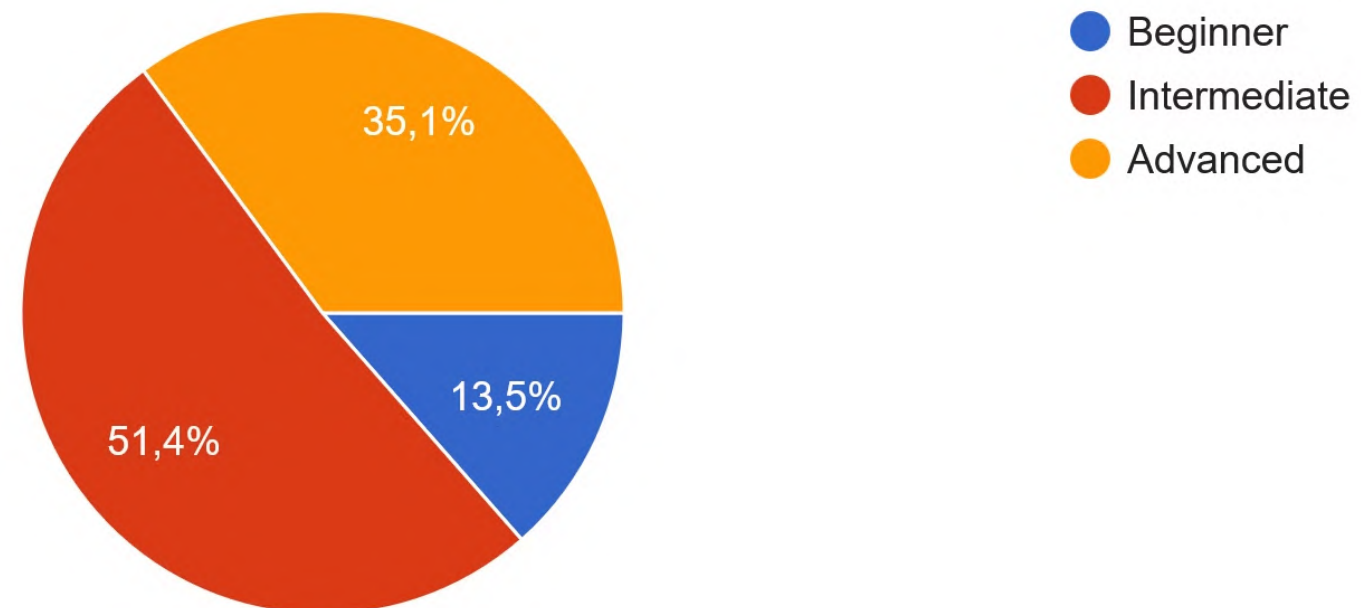
- fully booked for us
 - Rooms & Cleaning
 - Wifi: BFS-Guest
 - Ski Garage
 - Breakfast 7:30-8:30 (start at 8:30)
 - Dinner 18:30
 - NO LUNCH (only on demand €)
 - Drinks - tap water
-
- What needs to be paid?

Questions?

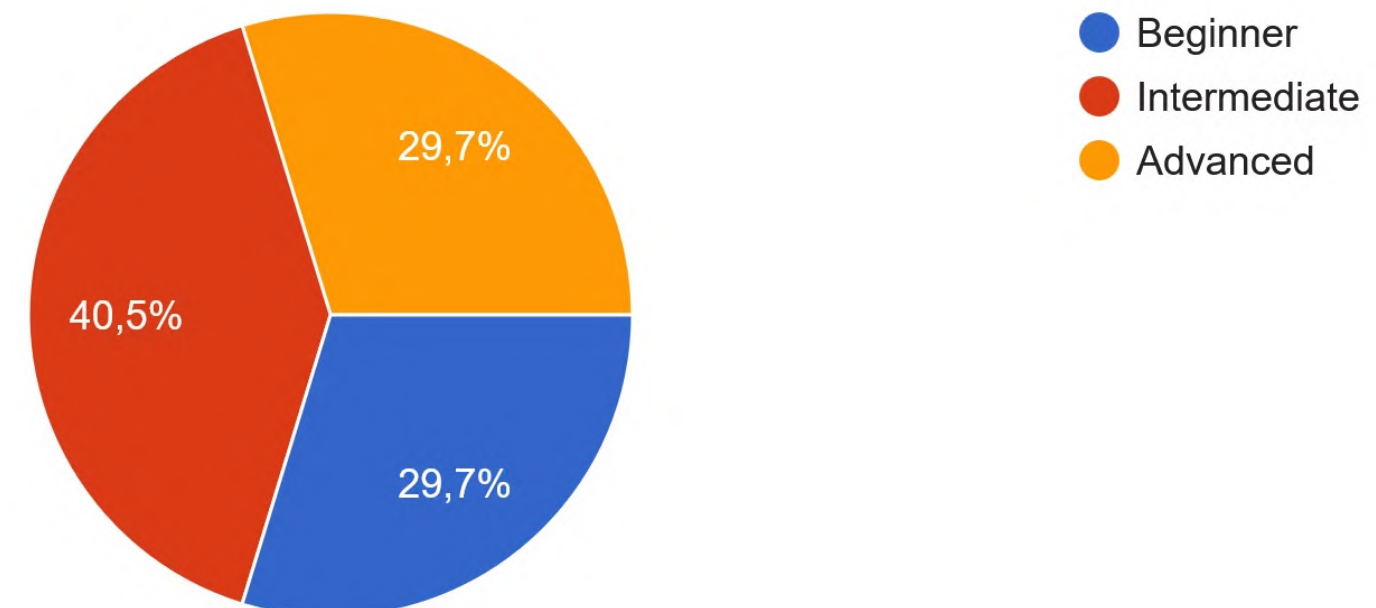
Statistics



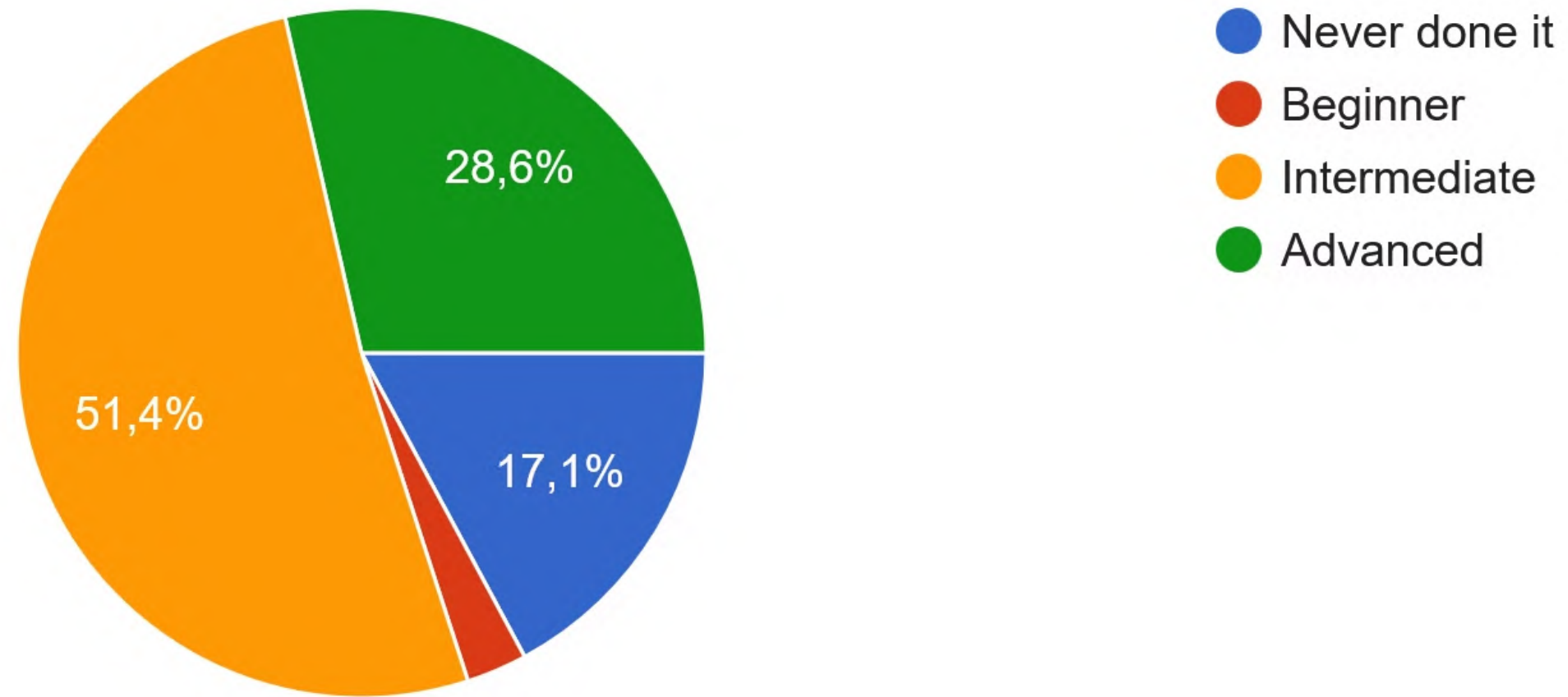
your level of Python programming



your level of ObsPy experience



Statistics - Wintersports



Wintersports - Safety First!



Workshop Agenda

	Morning 8:30-11	11-15	Afternoon 15-18	Evening
So		W I N T E R S P O R T S	Participants arrive Icebreaker 18:00, Restaurant Ground Floor	Introduction round, software setup, open fire
Mo	Workshop Welcome <i>Igel</i> Introduction to <u>Cryoseismology</u> <i>Fichtner/Walter</i>		<u>MSNoise&SeismoRMS</u> <u>Geopsy-HVSR on Greenland Darkzone data</u> + Alaska permafrost <i>Lecocq, Van Noten</i>	
Tu	From standard processing, wavefield gradients to migration. Analysis of <u>Grenzgletscher</u> and Greenland data <i>Wassermann</i>		Combining seismology and <u>physical simulation</u> of iceberg calving <i>Mangeney</i>	Bogner Ski Night ("Fire and Ice", and other ski movies)
We	DAS Practical: icequakes wavefield simulation, icequake source inversion <i>Hudson</i>		Free afternoon, parallel slalom, ski hut	
Thu	Seismology on a frozen lake: <u>seismo-acoustic effects</u> <i>Schmelzbach</i>		Science elevator talks from participants	Mulled Wine outside, open fire, special dinner
Fri	Science elevator talks from participants, Loose Ends, Final Discussions, Workshop conclusion <i>All</i>		Participants leave	

$$^2 = a^2 + \beta^2 \quad - \quad \frac{\partial a}{2 \cdot N} \approx 0.1$$

$$z = \frac{\sigma_0}{2\omega} + \frac{k'}{2'}$$

[illegible]

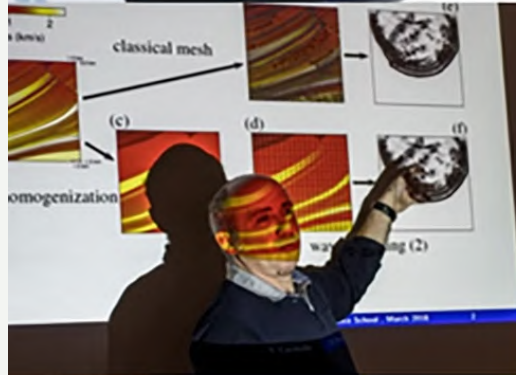
$$\frac{1}{\epsilon} (E = n \cdot T)$$

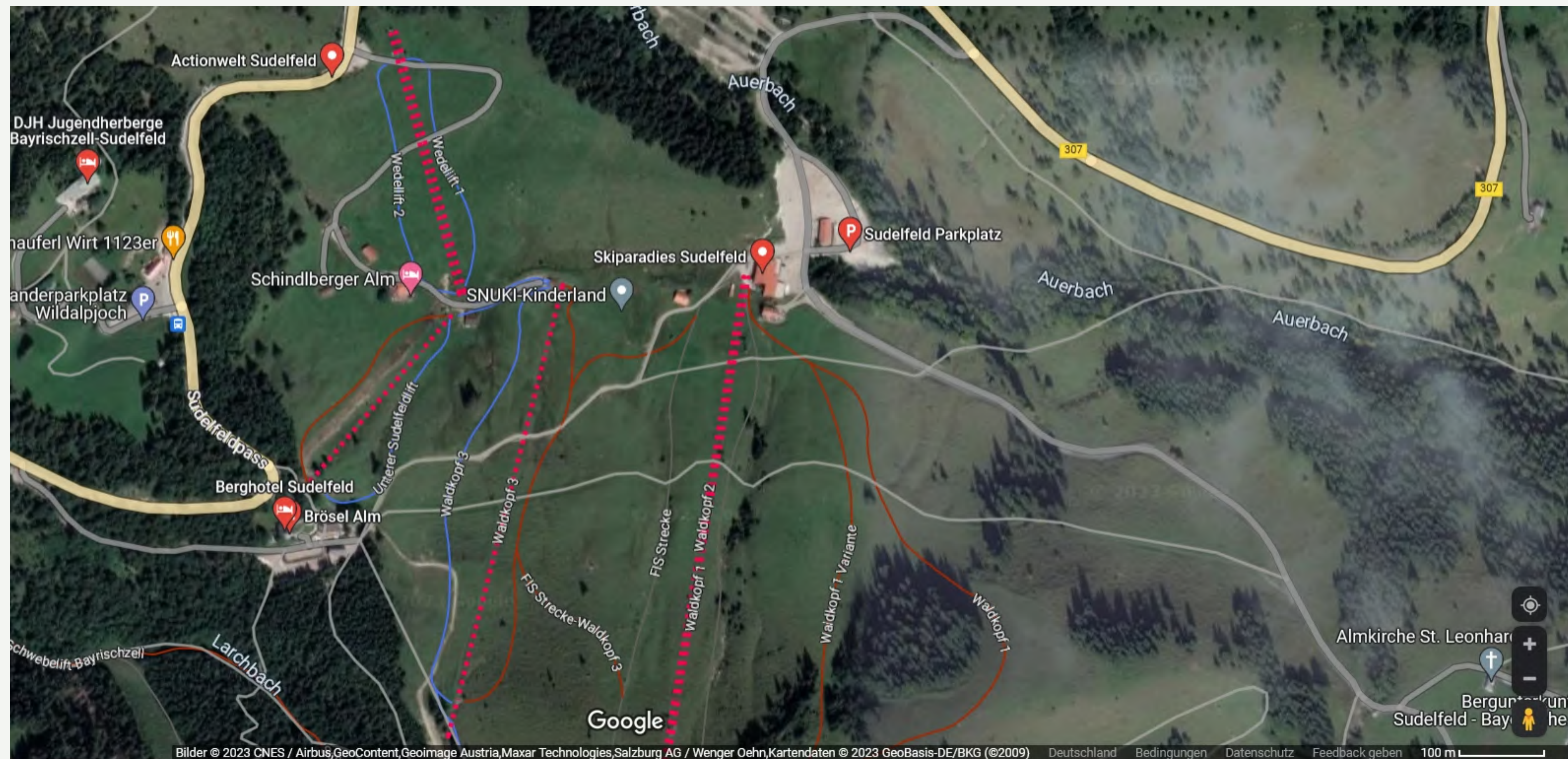




Try Pitch

20





Wintersport Information

- Ski Rental
- Ski pass
- Huts
- Beginners?
- Safety first
- Returning equipment
- Ski area
- Sun screen! (50+)



Gültigkeit	Erwachsene ab 19 Jahre	Senioren ab 65 Jahre	Jugendliche 16-18 Jahre Studenten* Azubis**	Kinder 6-15 Jahre
------------	---------------------------	----------------------------	--	-------------------------

Tages- und Halbtageskipässe				
Tageskipass	53,00 €	50,00 €	46,00 €	26,00 €
Vormittagskarte bis 13 Uhr	46,00 €	45,00 €	38,00 €	20,00 €
Nachmittagskarte ab 12 Uhr	43,00 €	42,00 €	36,00 €	19,00 €
Happy-Hour-Skipass täglich ab 14.30 Uhr	23,00 €	23,00 €	23,00 €	10,00 €

Stundenkarten				
3 Stunden	42,00 €	40,00 €	35,00 €	20,00 €
4 Stunden	47,00 €	45,00 €	40,00 €	21,00 €
Mittwochskracher (Tagesskipass) Mittwochs ab 07.01.2026 (außer Aschermittwoch)	37,00 €	37,00 €	37,00 €	26,00

Ski Prices 2026

- Cheapest strategy - Adult/Student**
- ✓

Monday → 3h ticket: 42€ / 35 €
- ✓

Tuesday → 3h ticket: 42€ / 35 €
- ✓

Wednesday → promo day pass 37€ / 31 €
- ✓

Thursday → 3h ticket: 42€ / 35 €
- Total ≈ 163 € / 137€

- Punktekarte**
- ✓

20 points = 16 €
- ✓

Chairlift ~8-9-10 points
- ✓

Lift ~2-3 points
- Total ≈ depends on you

Lifte und Bahnen		Punkte
A	Schwebelift Bayrischzell	9
B	Kitzlahner 4er-Sesselbahn	9
C	Waldkopflift 6er-Sesselbahn	8
D	Sudelfeldkopf 8er-Sesselbahn	10
E	Schöngrat 3er-Sesselbahn	9
F	Rosengassenlift	9
G	Oswaldlift	2
H	Vogelsanglift	3
I	Wallerlift Monto Gogo	2
J	Mittleres-Sudelfeld-Lift	3
K	Plattenlift	2
L	Wedellift	2
M	Hotellift	2
N	Kinderland	

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- Punktekarte**
- ✓ 20 points = 16 €
 - ✓ Chairlift ~8-9-10 points
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160€ for 4 days

138€ for 4 days

(student ID)

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Logistic Monday PLEASE REMEMBER

7:30 Breakfast

8:30 - 11:00 Lecture

11:00

Good skiers: Walk to Rental Ski place, leave shoes in “SKIENCE” Bag,
come

back to hotel with skis (we bring shoes back)

Beginners: Meet in Parking 11:10, van leaves 11:15 to ski rental place, ski
lessons to 14:30, van back to hotel 14:30

15:00 Lectures

Communication: Mattermost

Practicals: SKIENCE26 conda environment

<https://github.com/heinerigel/Skience2026>

download skience26.yaml

run:

conda env create -f skience26.yaml

GOOD NIGHT



Seismic Tracking 2026 Planning



CODES for practicals!

- open a console (bash, console, cmd, etc) - and run the command
- *git clone <https://github.com/heinerigel/skience2026.git>*
- it'll create a skience26 folder structure with the codes for the practicals
- if you already did it before *now*, remove the skience26 folder and redo it (or you might need to run complicated git commands...)

DATA for practicals!

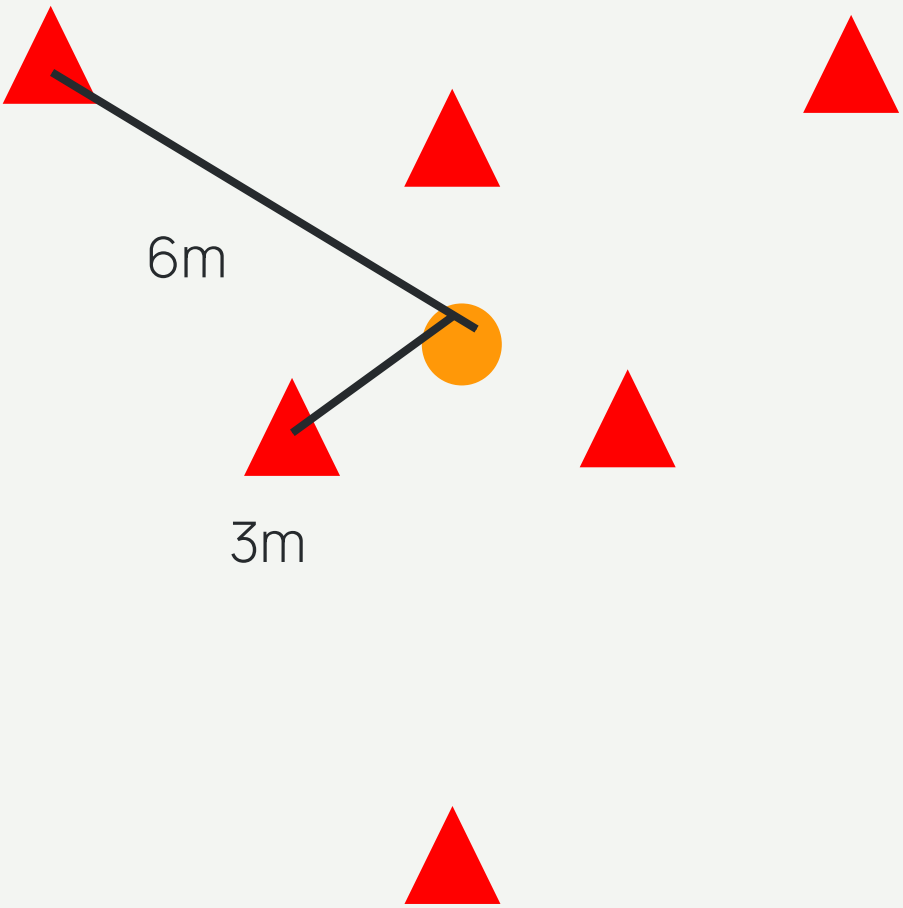
- Two USB disk are going around
- copy SKIENCE26_DATA to your computer in any local folder: **NOT IN THE skience26 git FOLDER!**

Nodal Experiment

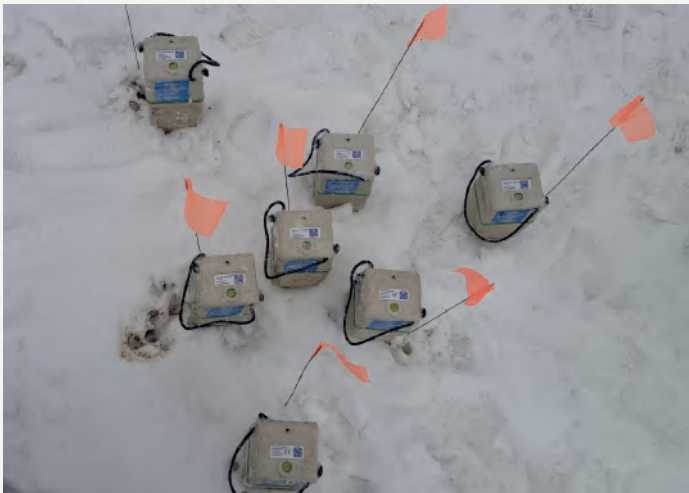


ADR Layout

- 2 arrays
- 2 openings
- 7 3C 5Hz smartSolo's

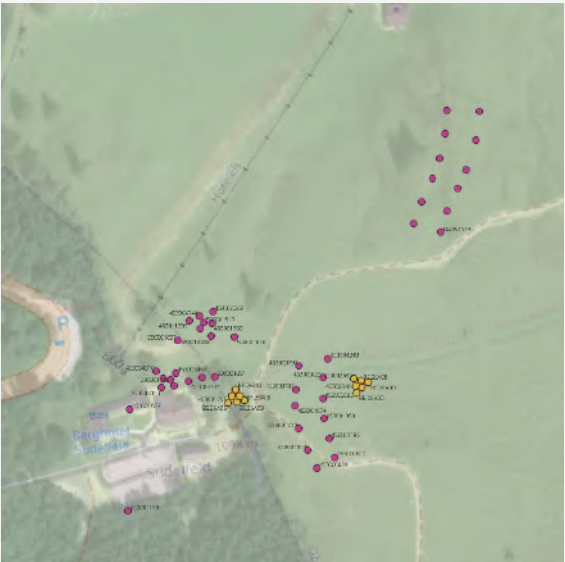


Skience26 Nodal Experiment



Illustration, not the real positions

Make Heiner happy map



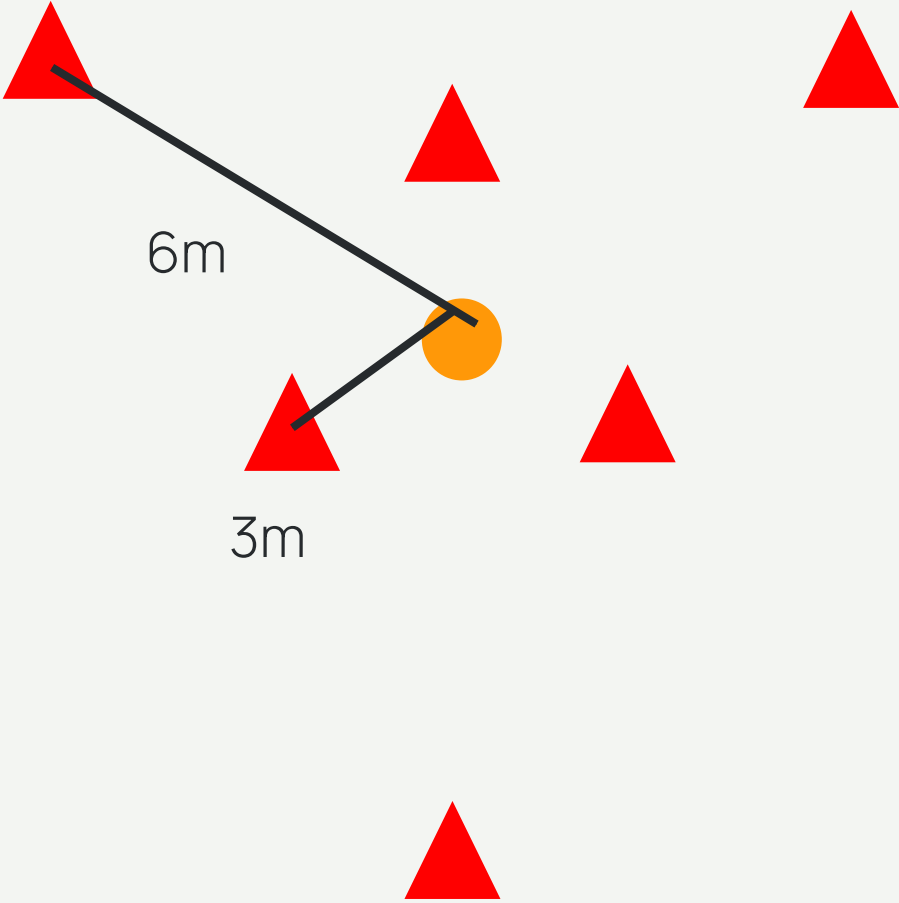
dGPS positions

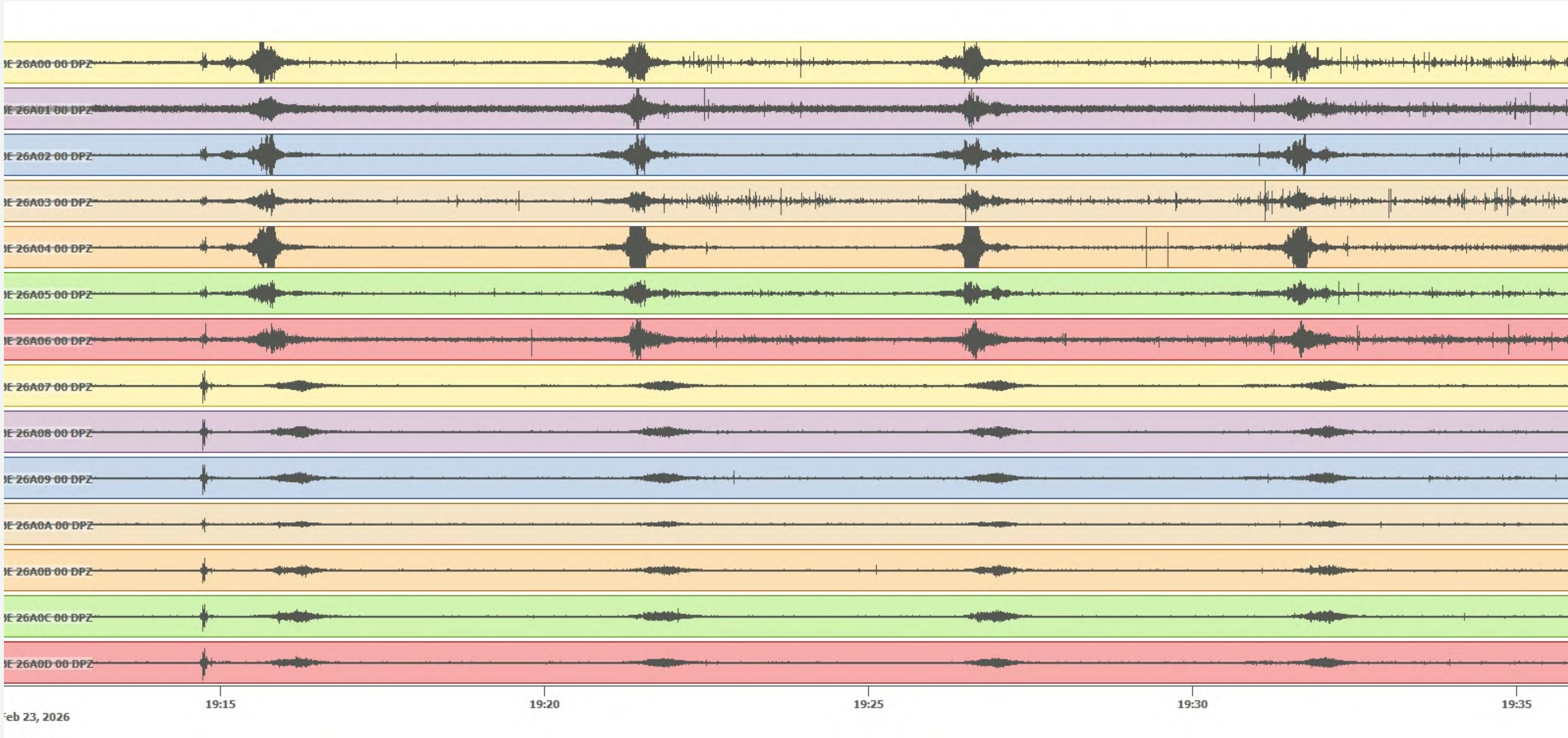


Try Pitch

ADR Layout

- 2 arrays
- 2 openings
- 7 3C 5Hz smartSolo's





“CHALLENGE” for Tuesday Evening

Prize : Belgian Cookies

Use/copy the notebooks to:

Plot the temporal 1970s posters for all sensors

Analyze the PSDs and PPSDs

Run the HVSR analysis

Run the tracking ADR analysis

report progress on Mattermost :-)

Data: on the **USB disks**:

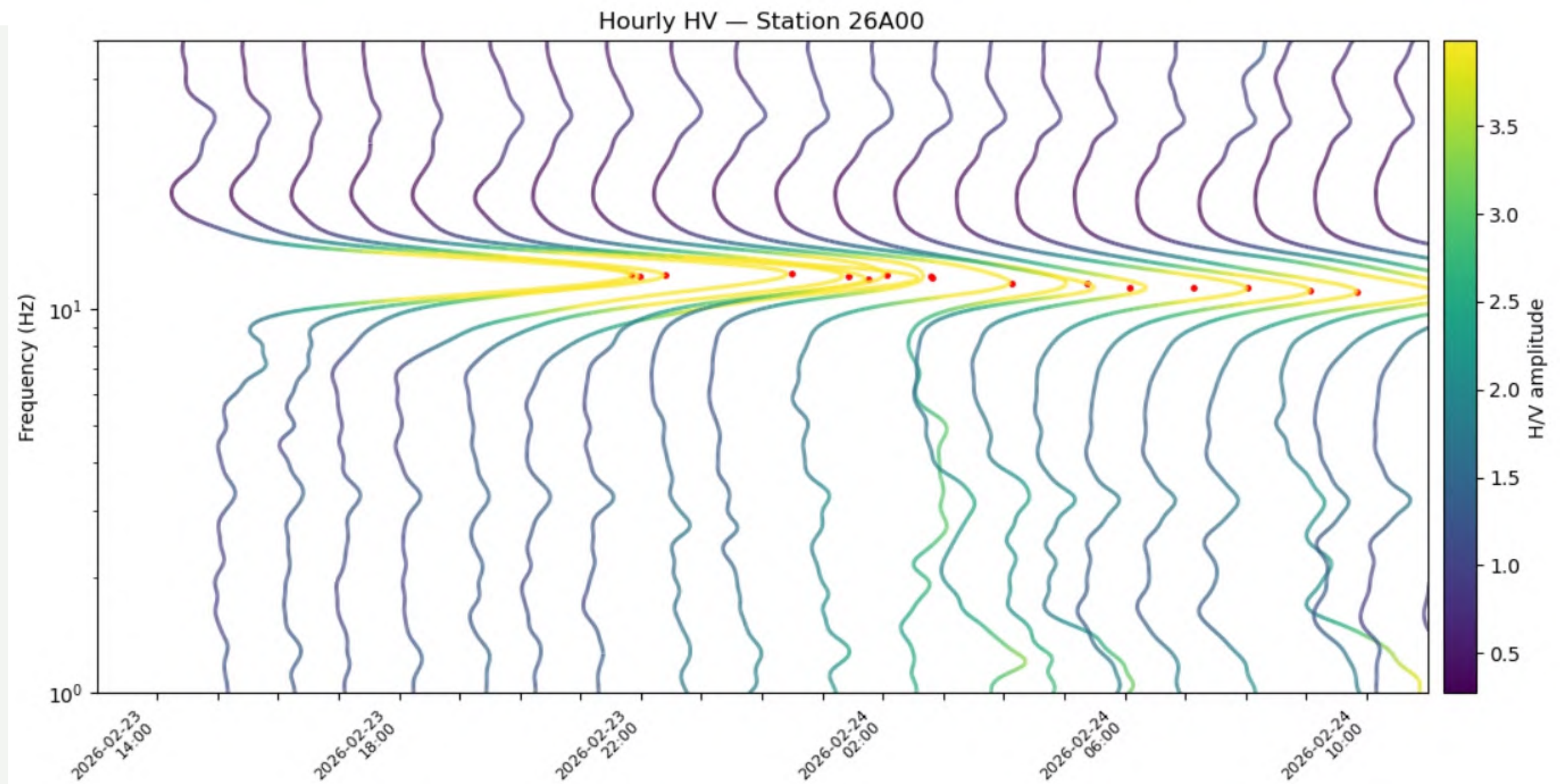
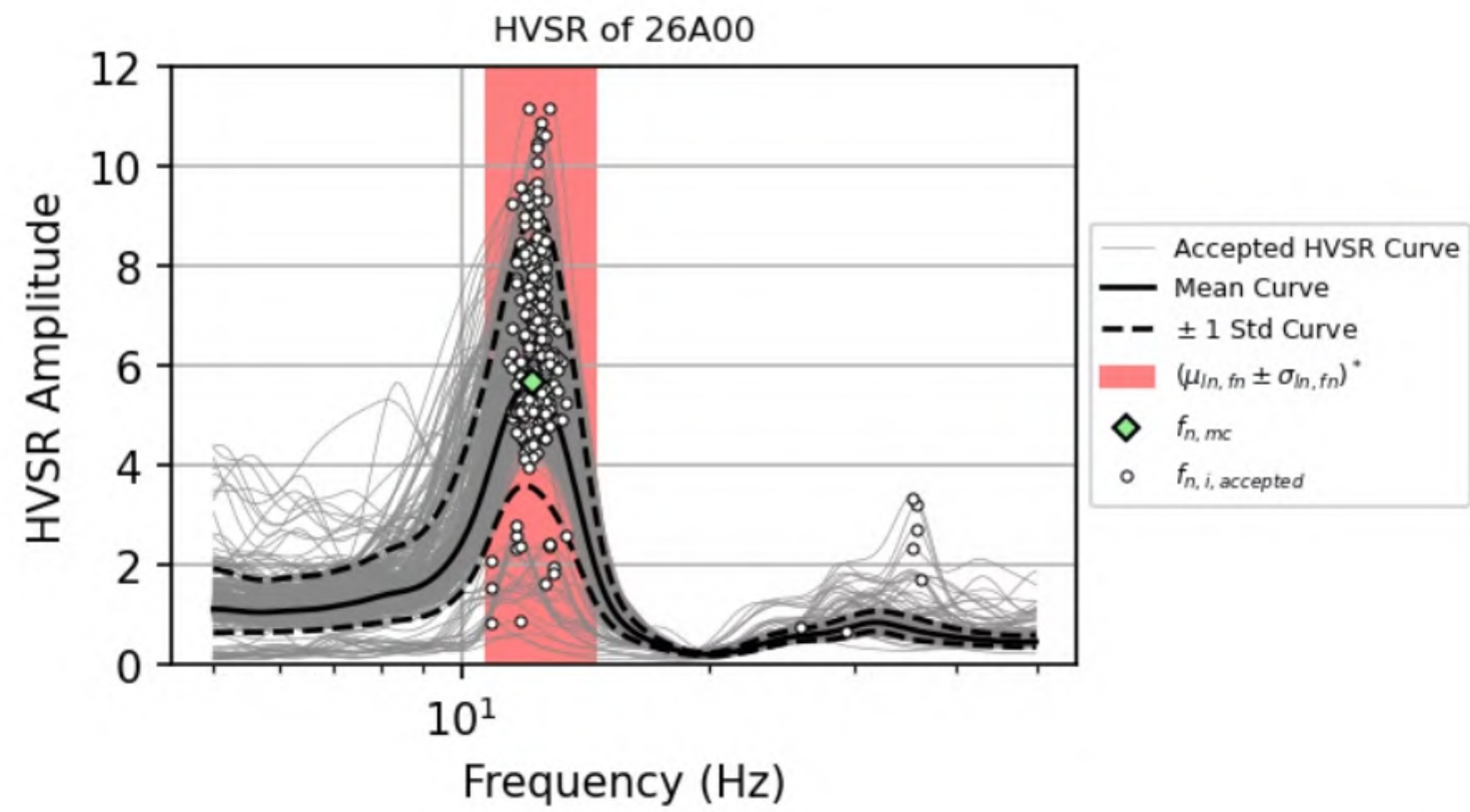
/SDS_nodes : the archive of the 14x3 components x2 days

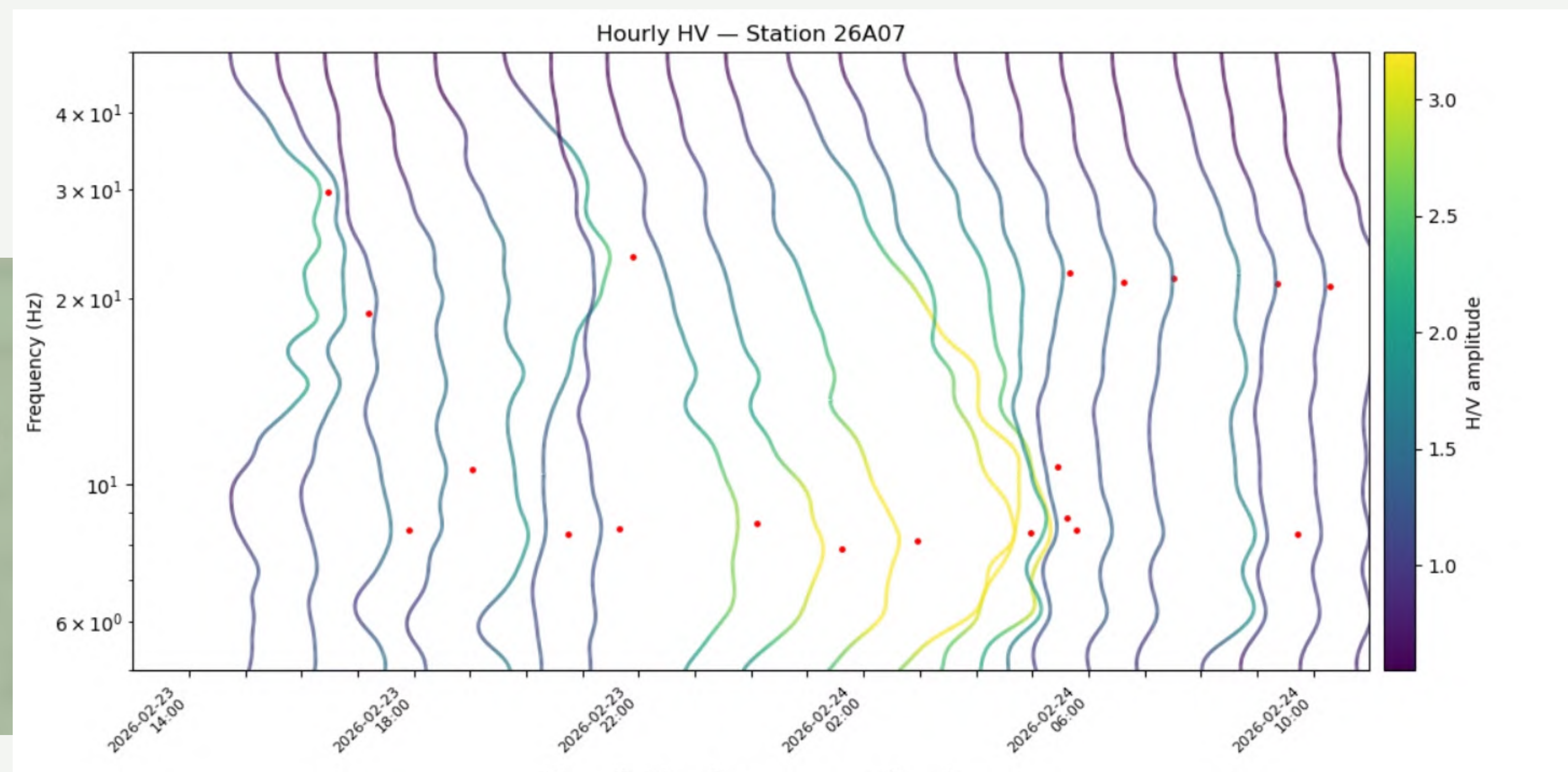
/stationxml_nodes : the smartsolo responses



0 30 m





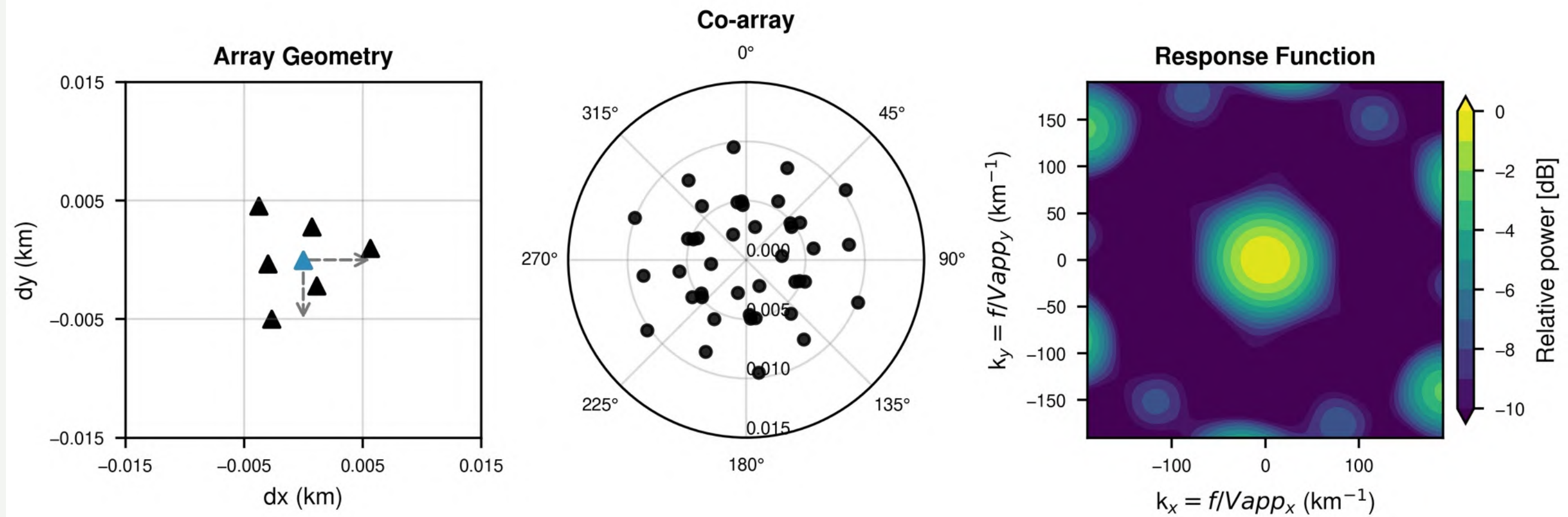
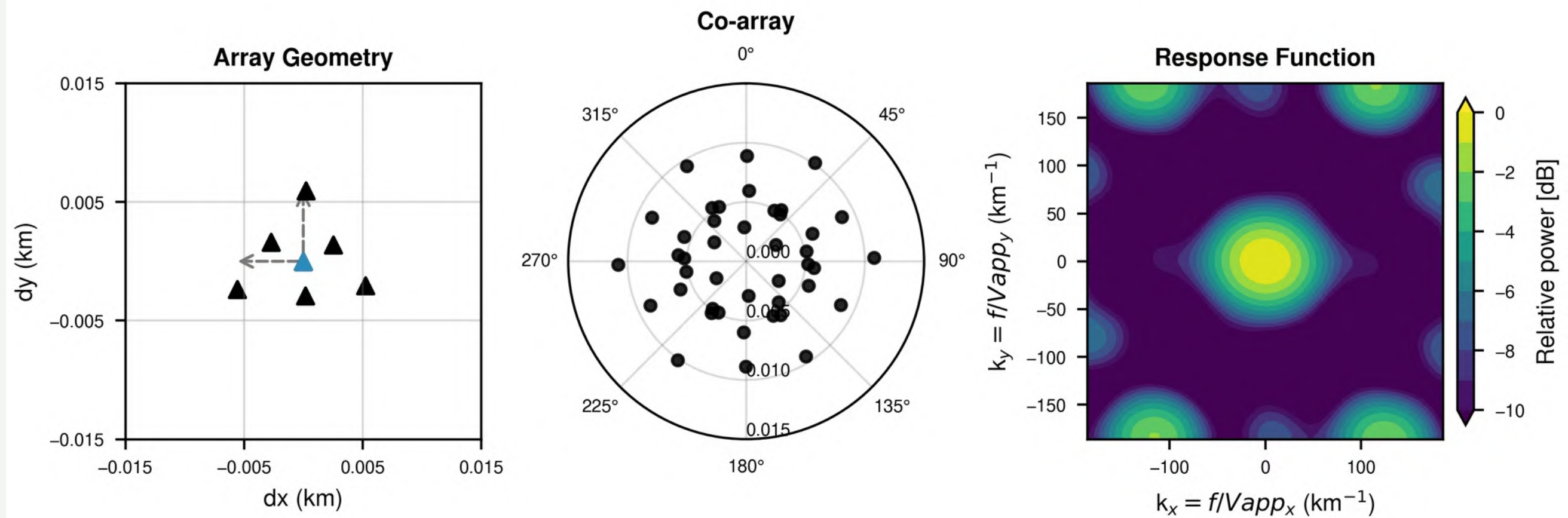


0 30 m

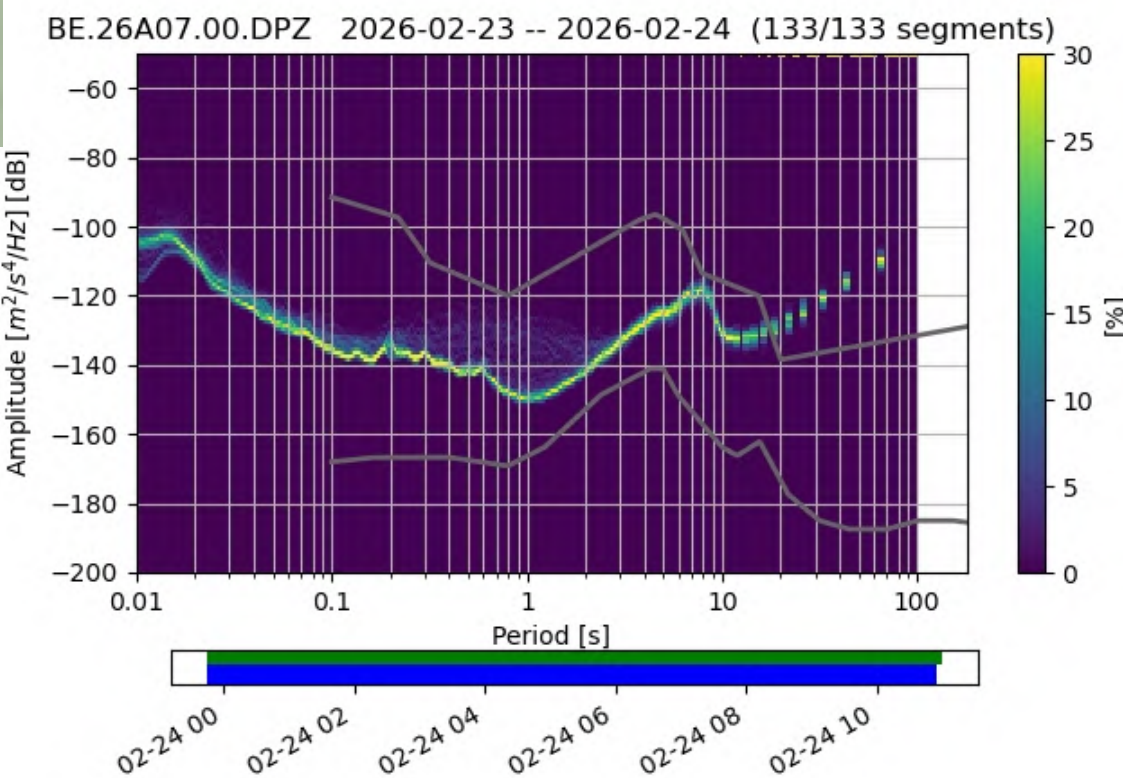
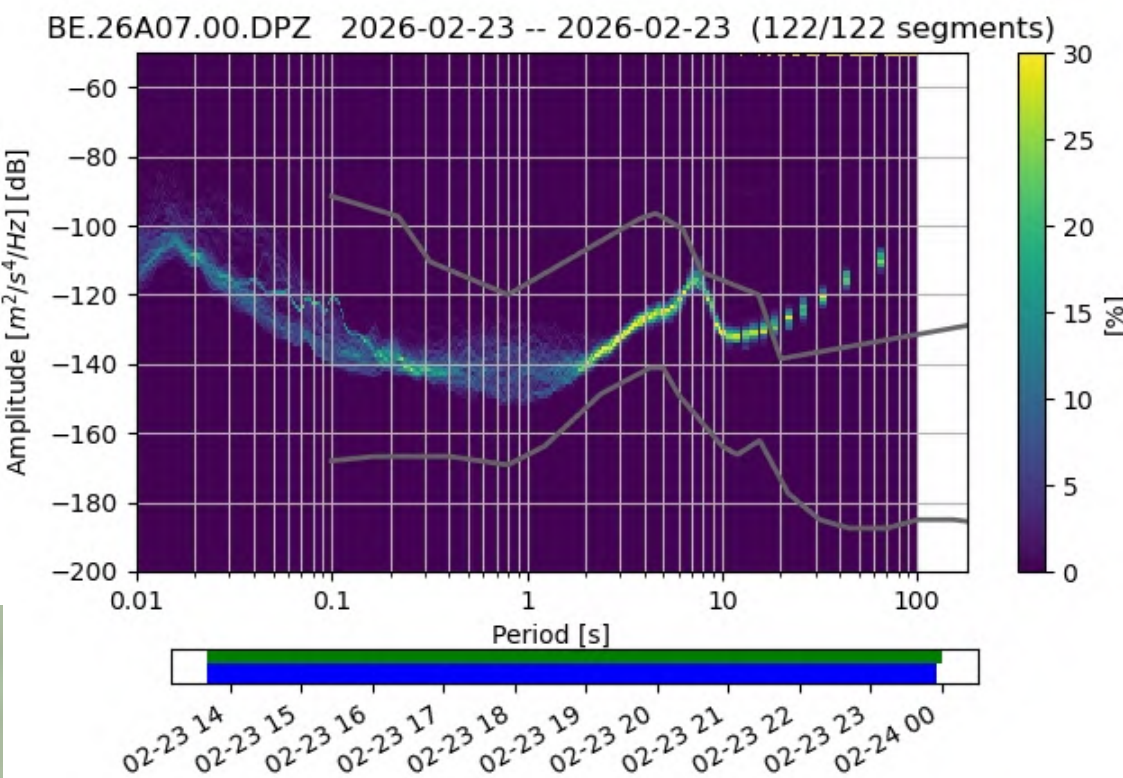
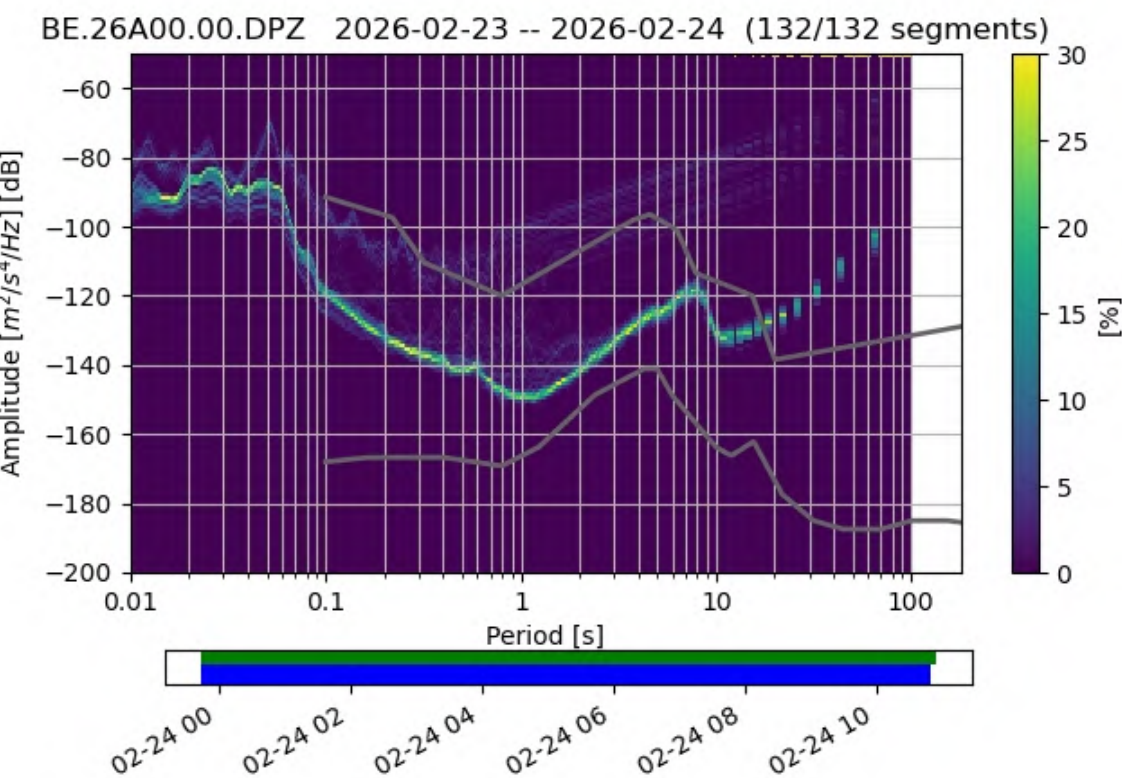
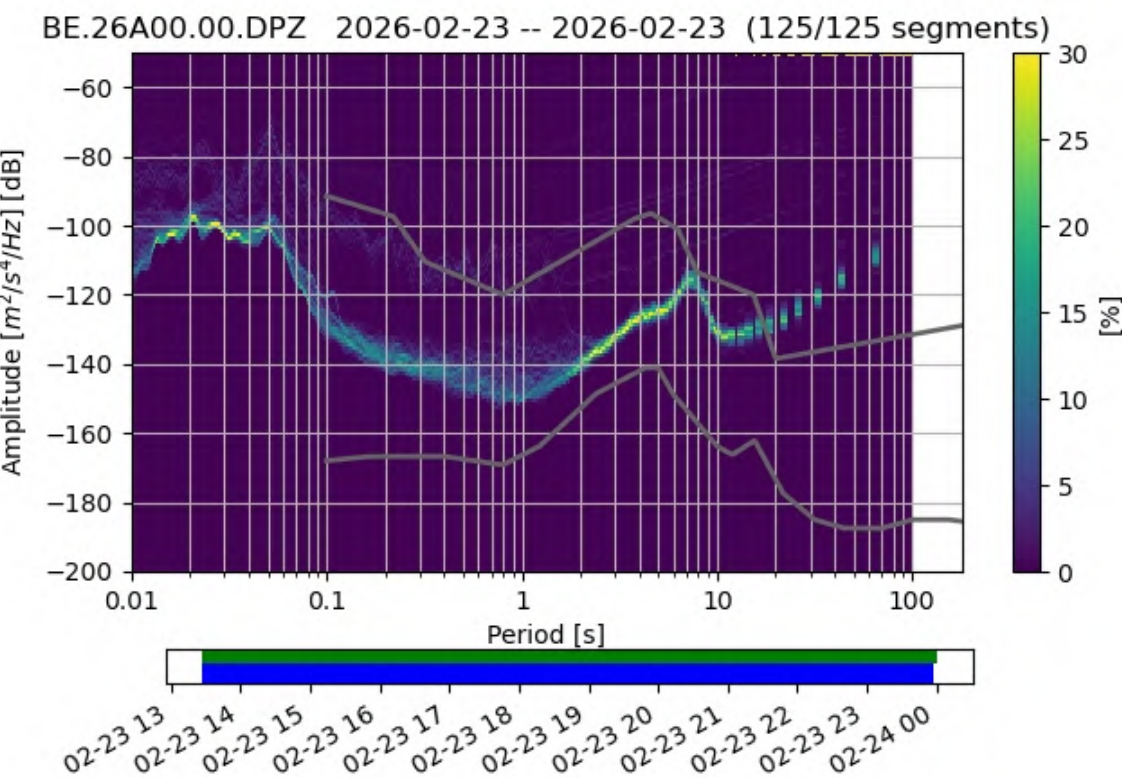


0 30 m

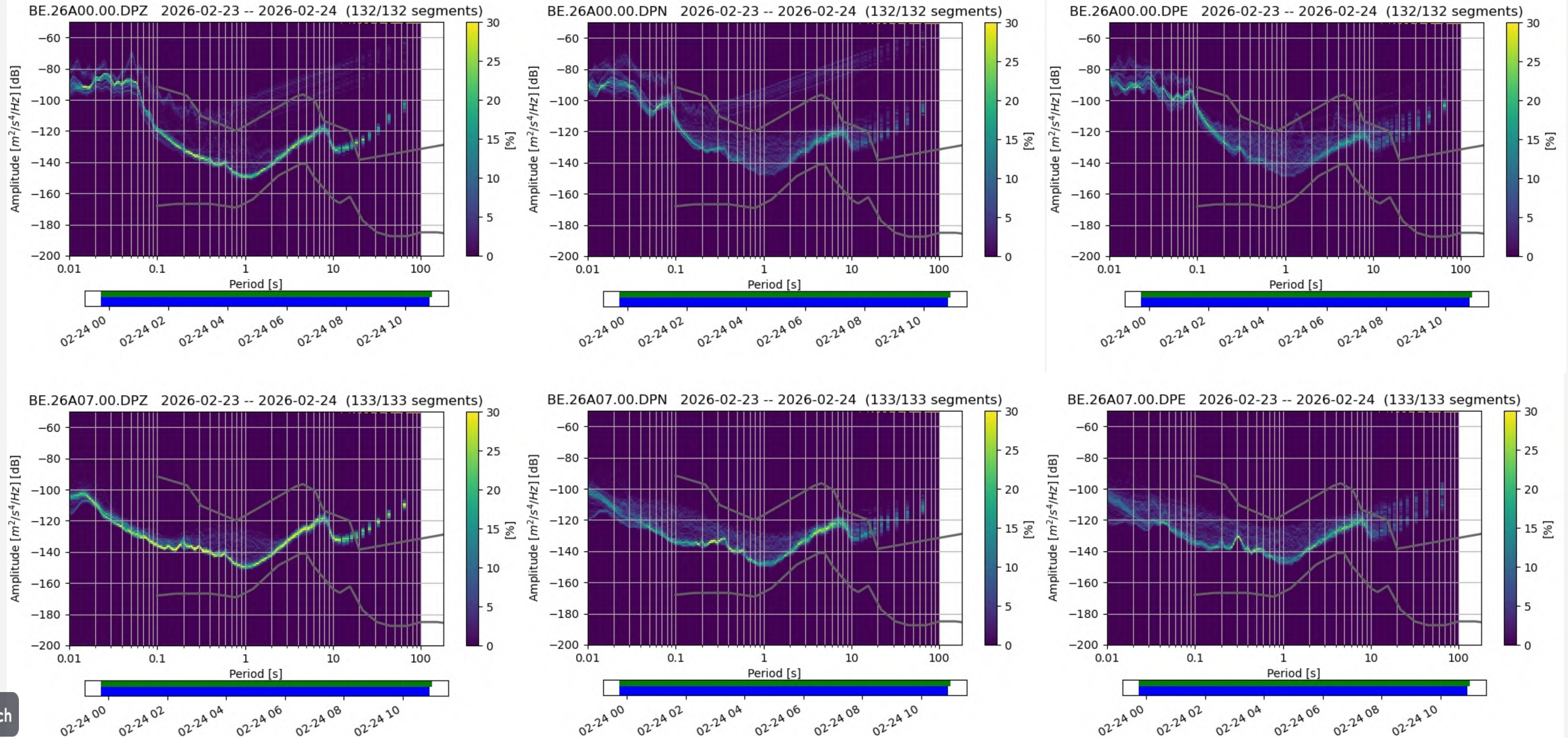




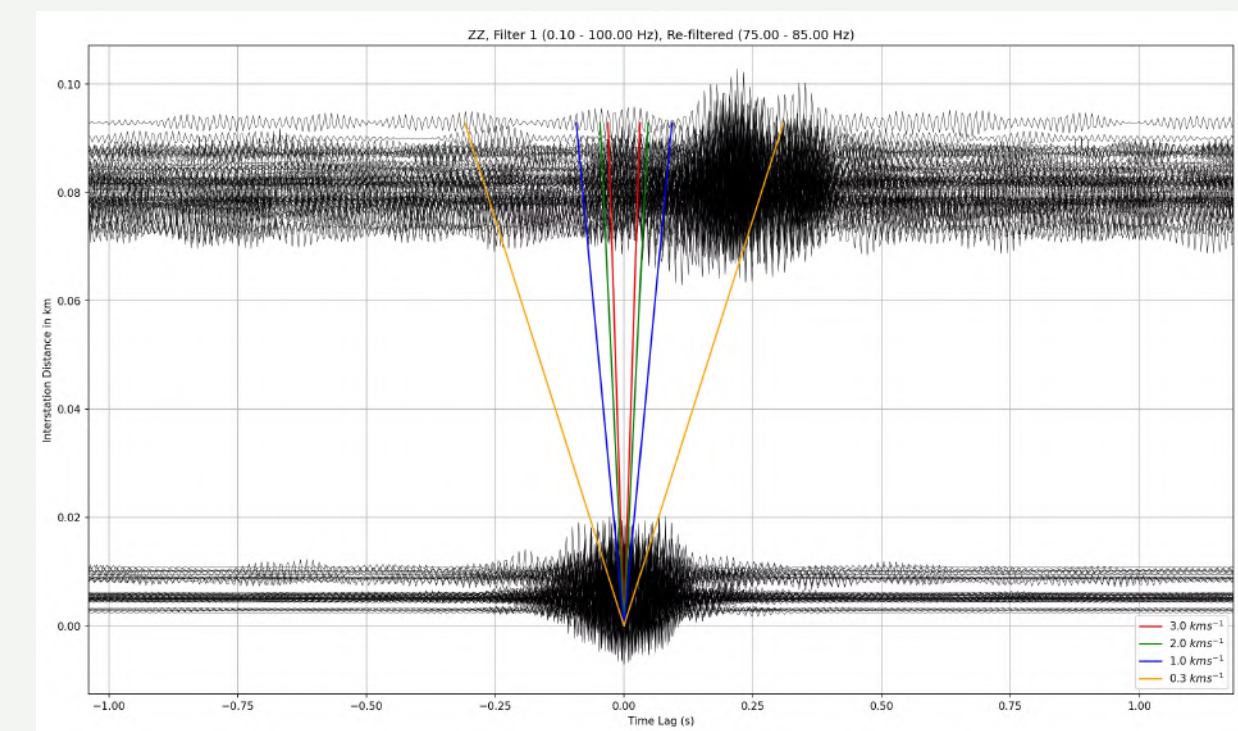
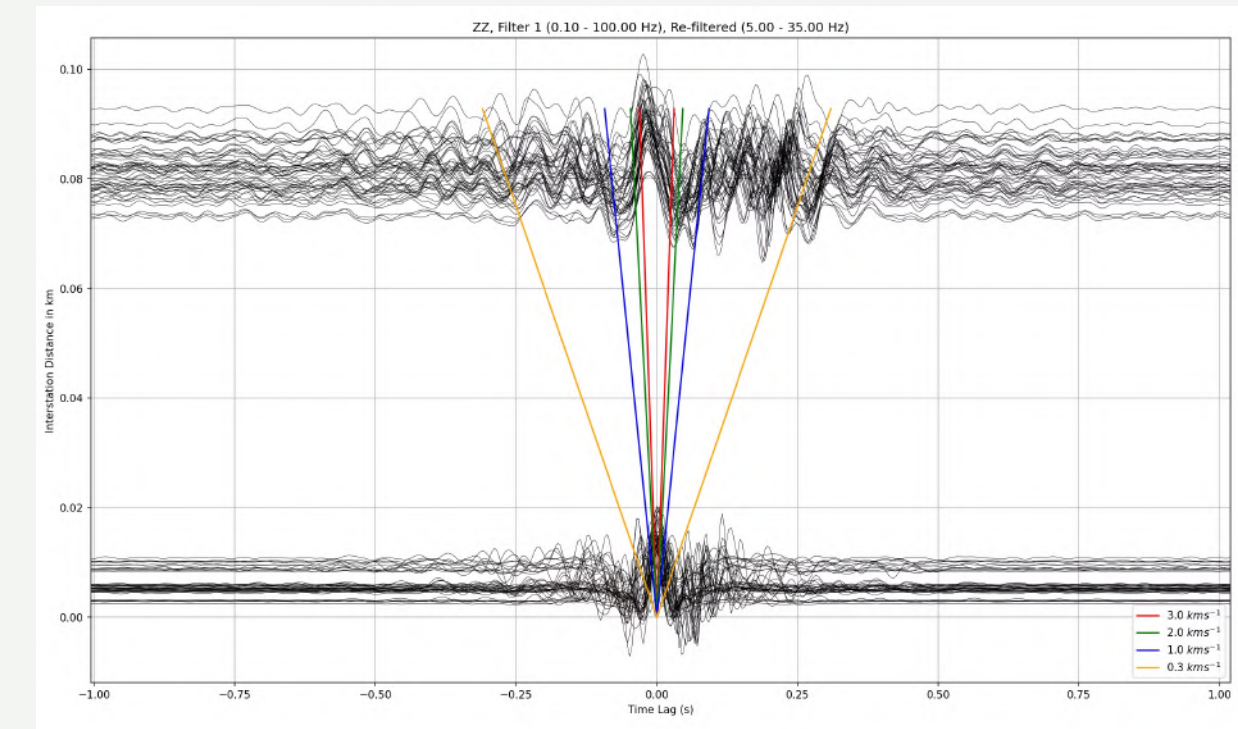
Noise levels



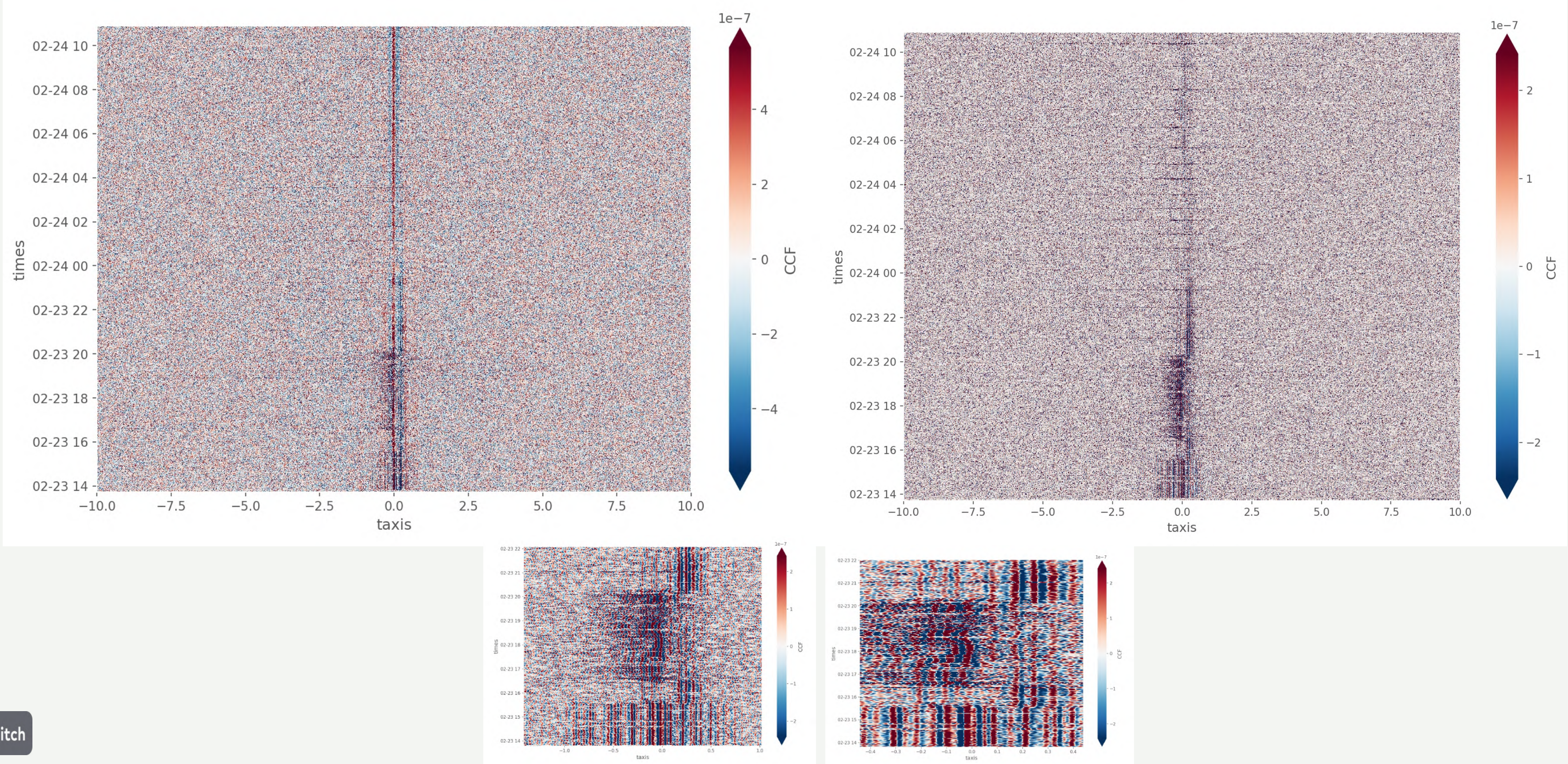
Noise levels



Cross-Correlations



Cross-Correlations



XX February 2023

Tracking Skiers 2025 pre-results

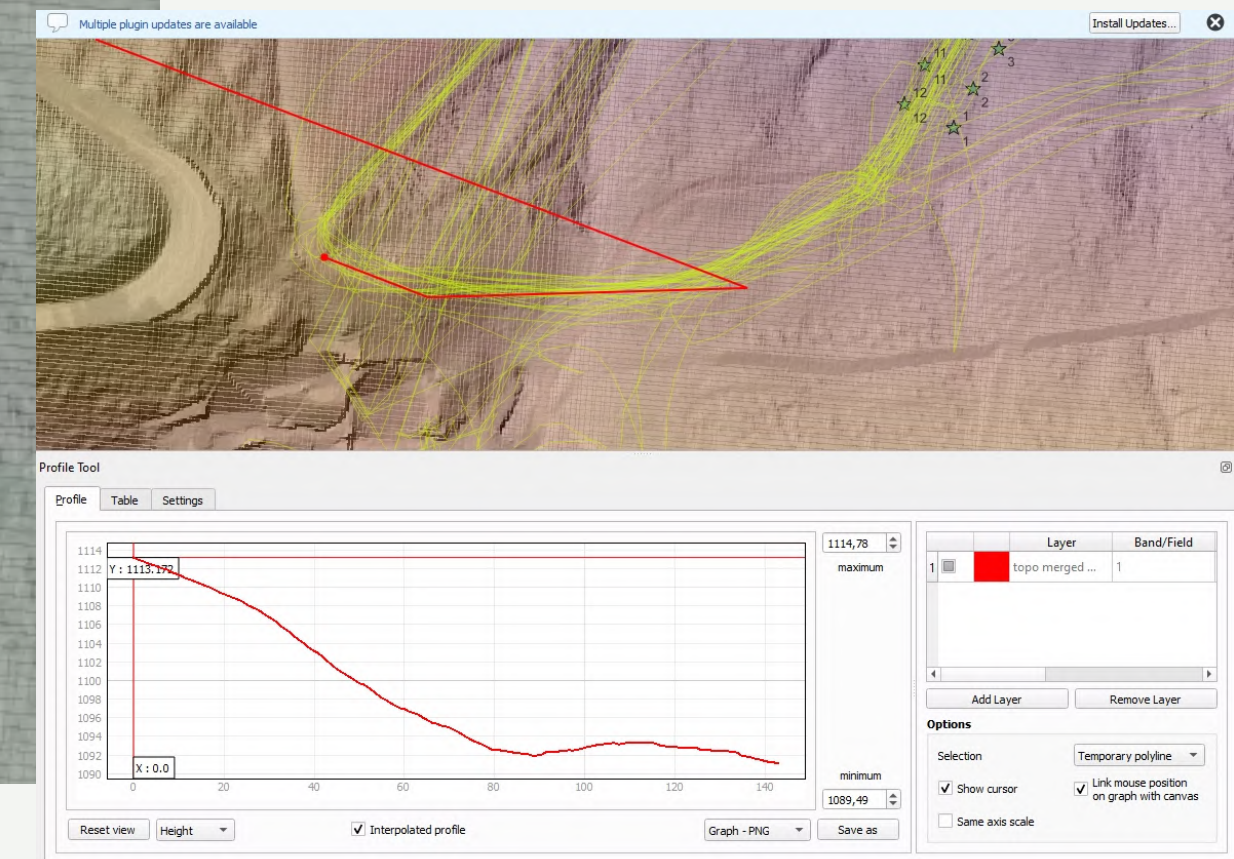
15th Munich Earth Science School
Environmental Seismology

Try Pitch



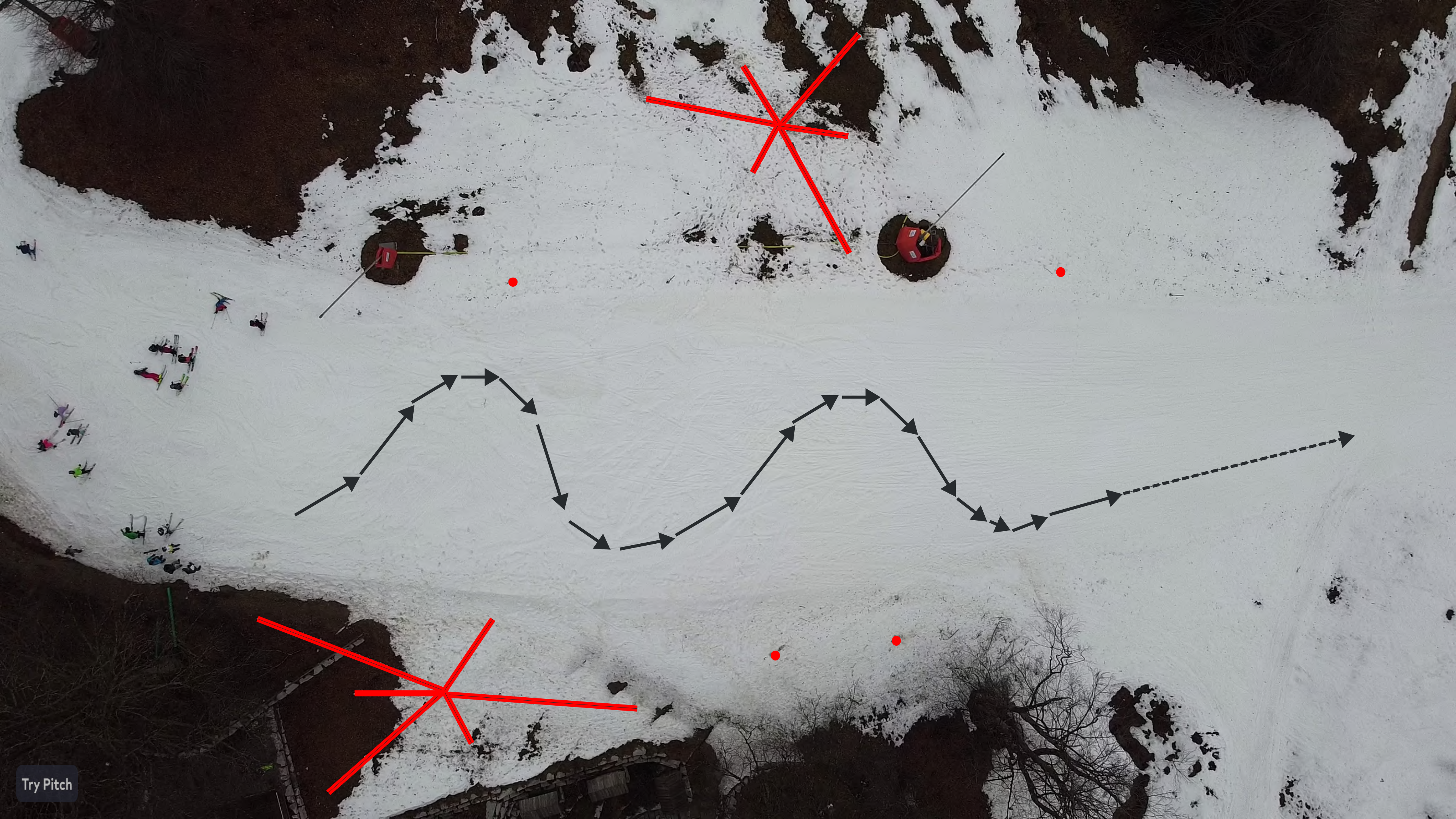
Track you ski
effort !

Strava
Garmin
Whatever







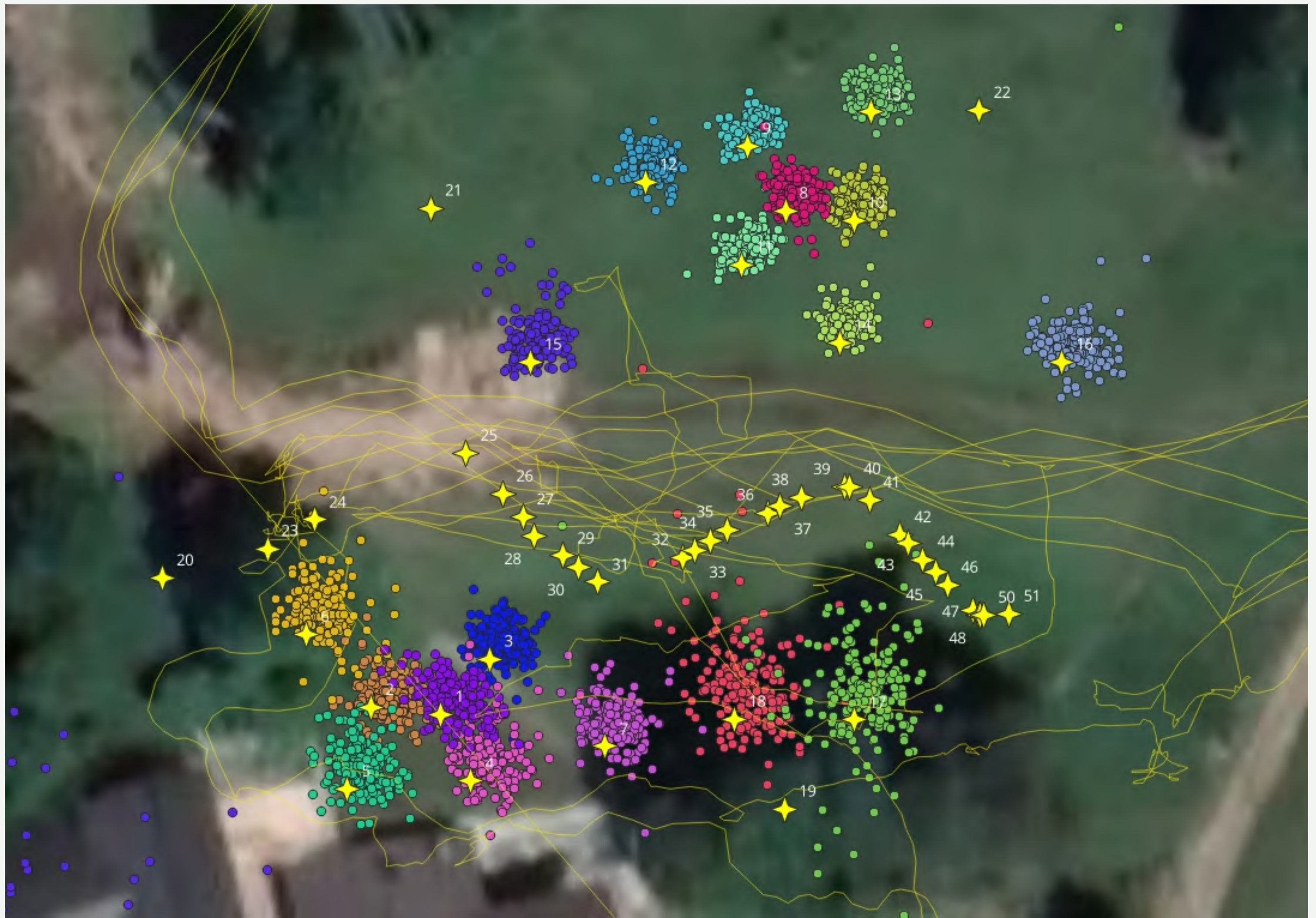


Tracking Skiers 2025 first results

Data will be shared:

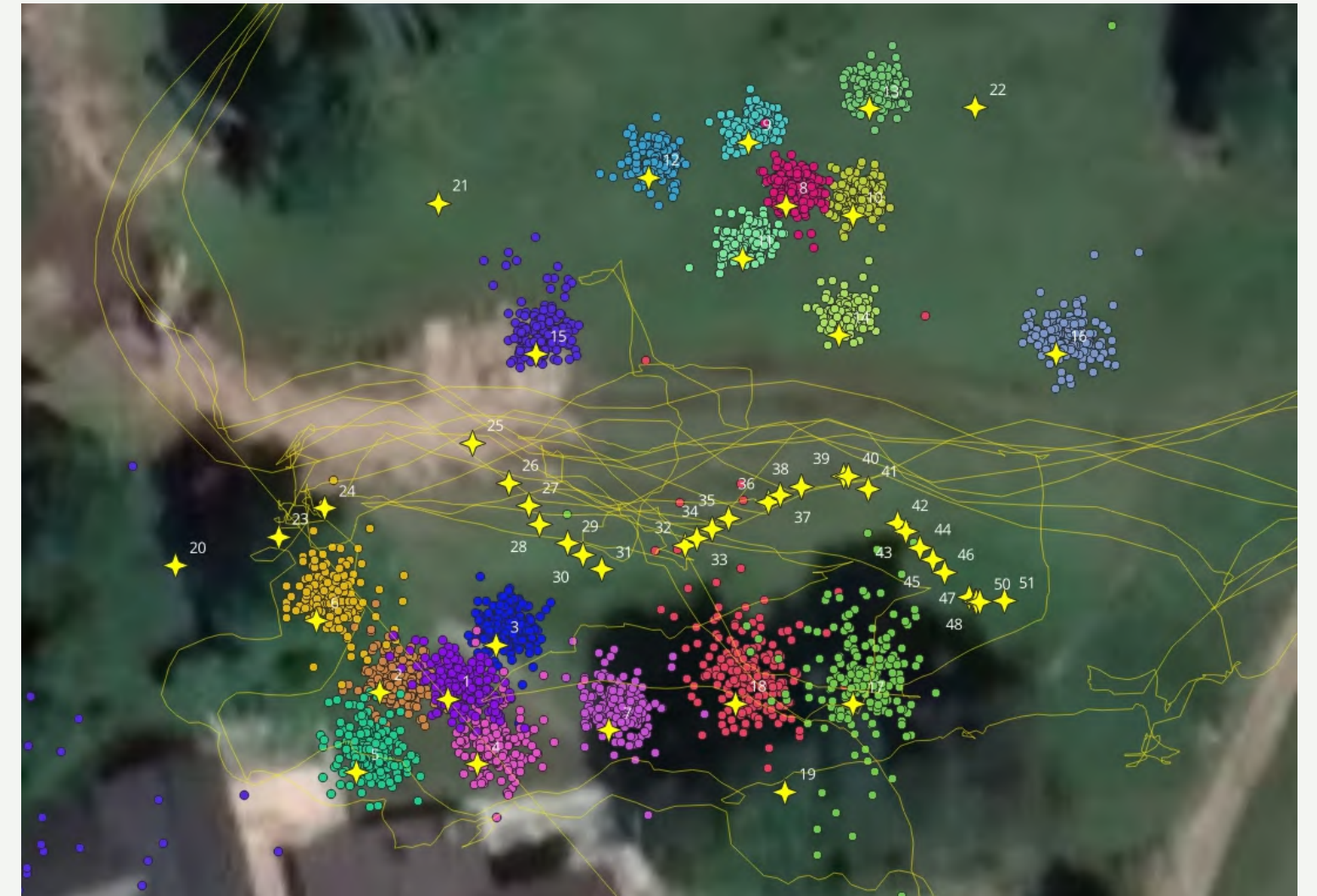
- SDS (4.4 GB - 500 Hz data)
- Response
- dGPS positions
- Logbook with timings
- Drone video footage
- other photos & videos







Decoupling due to melting?





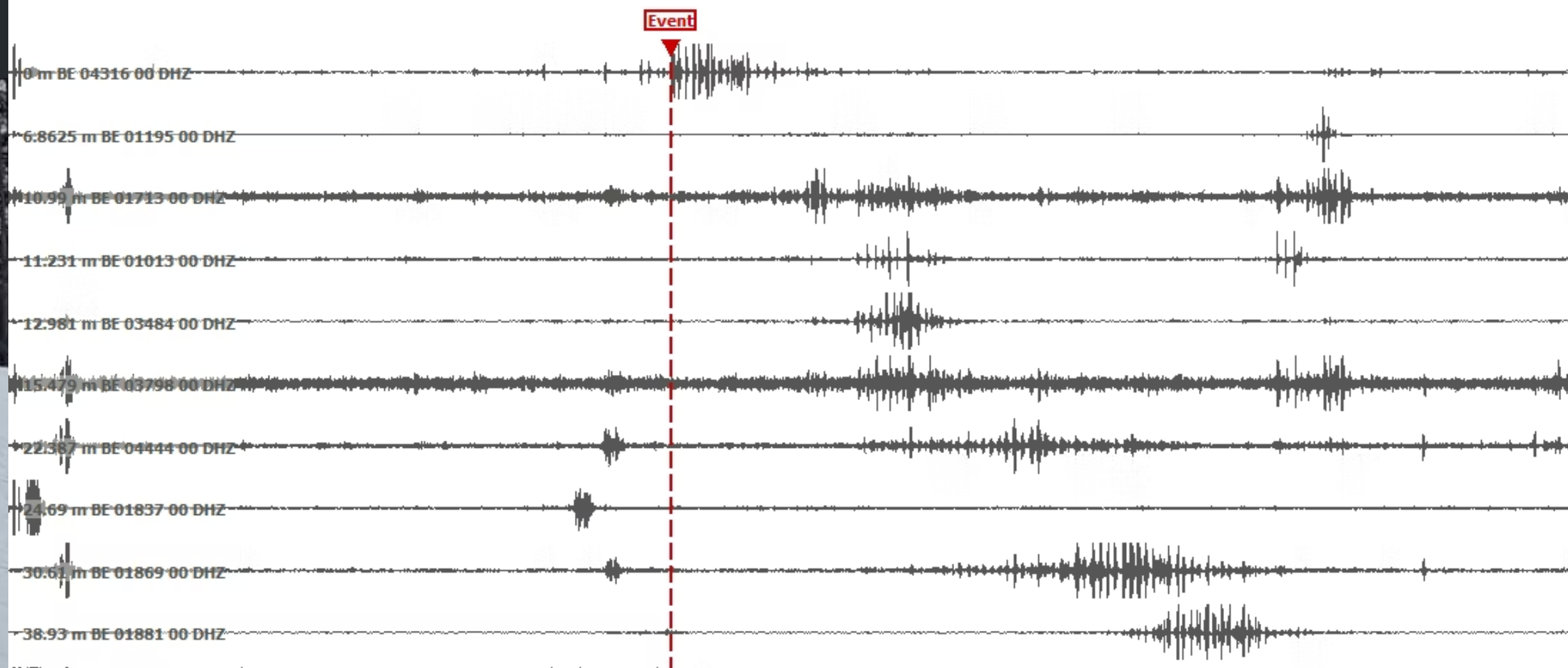




Heiner Test 1 Skience 2025

S Array

We see
Heiner!



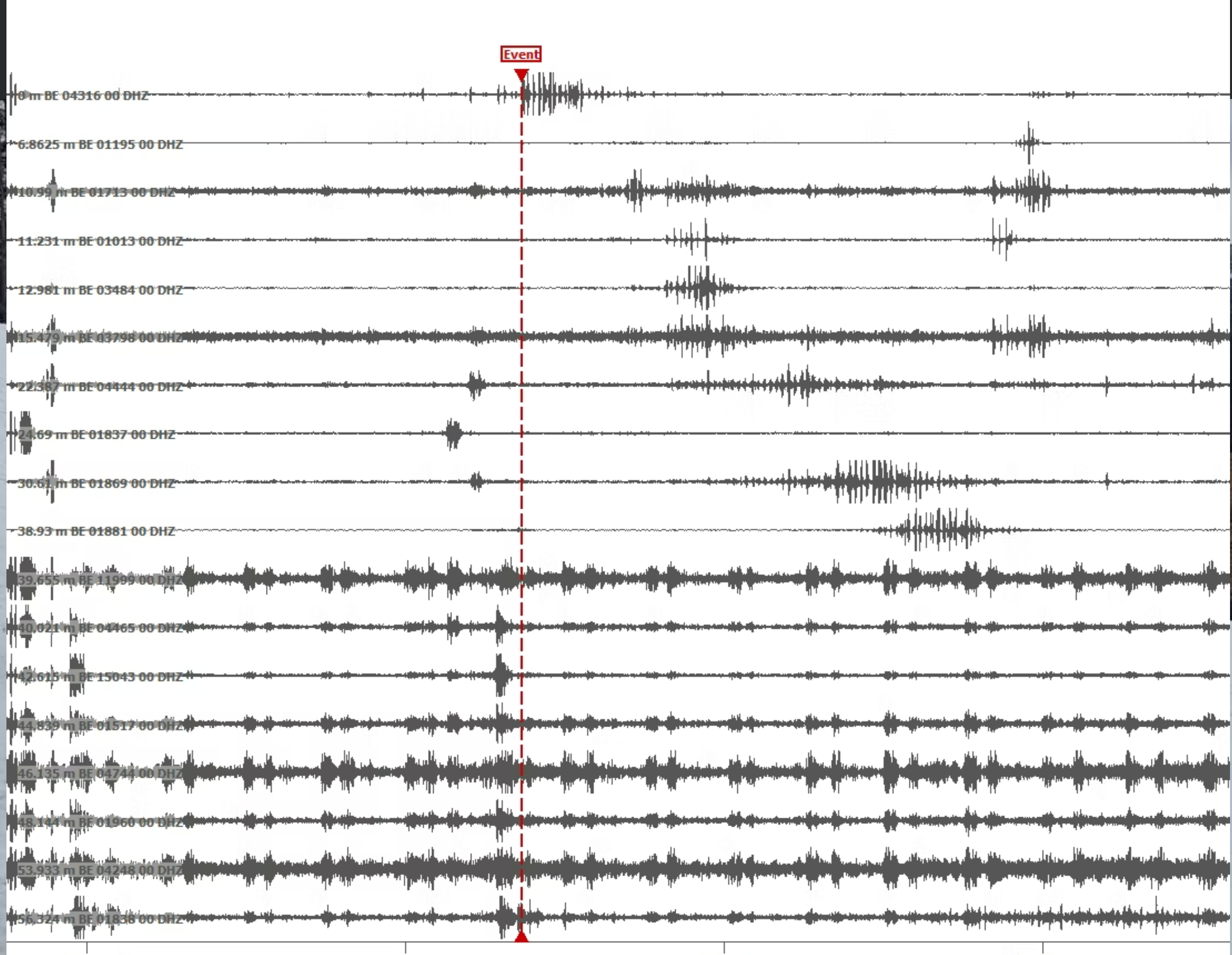
Heiner Test 1 Skience 2025

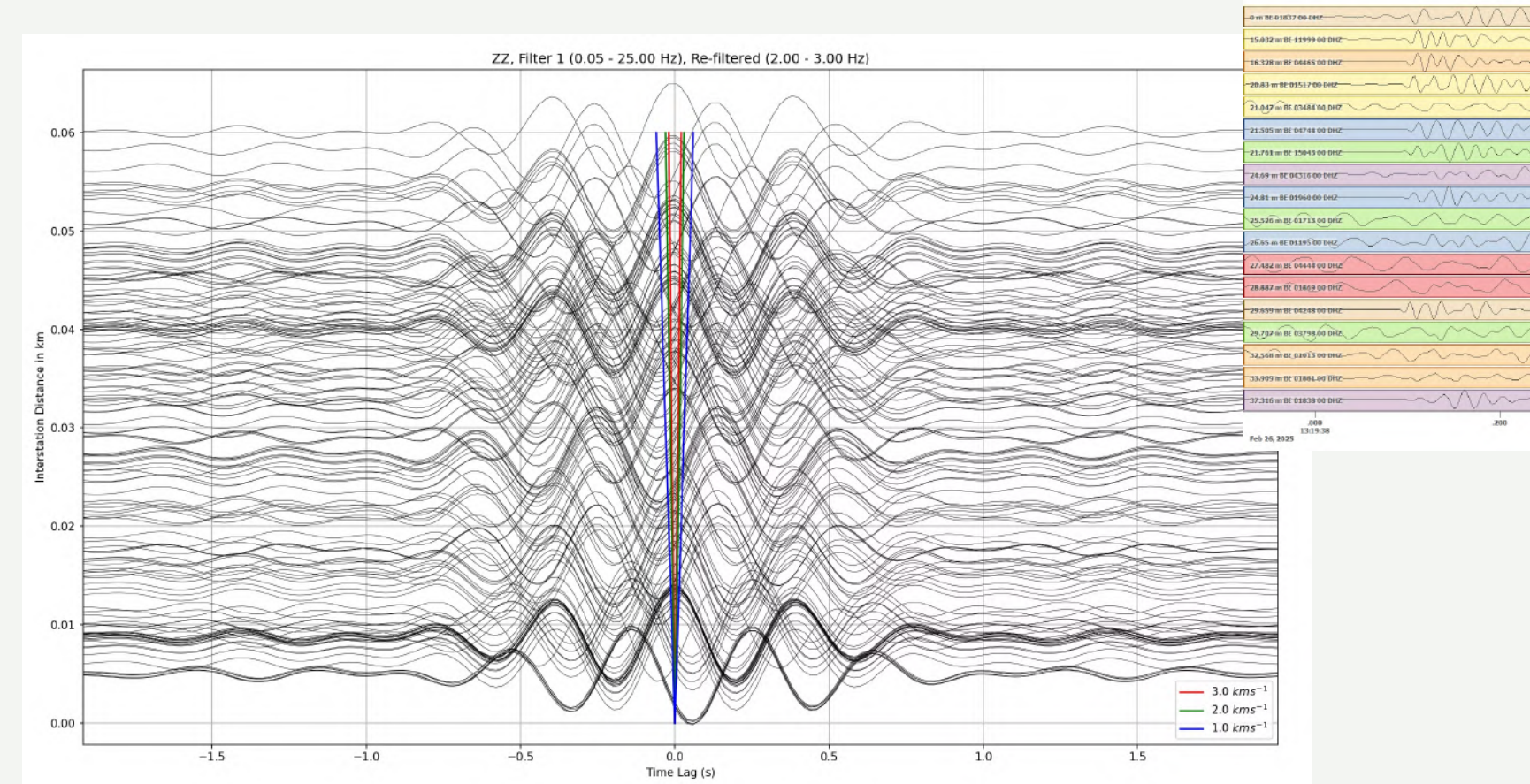
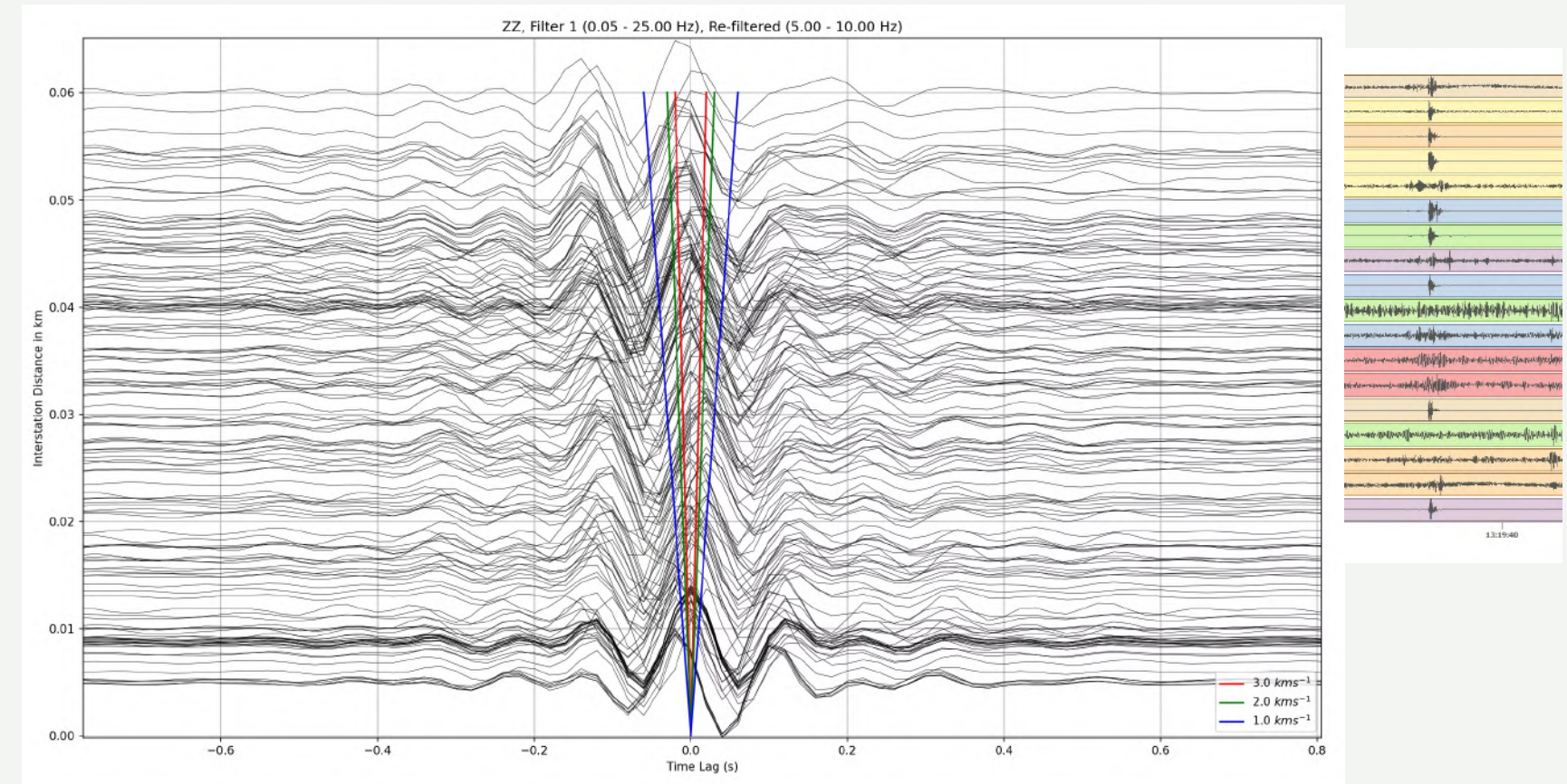
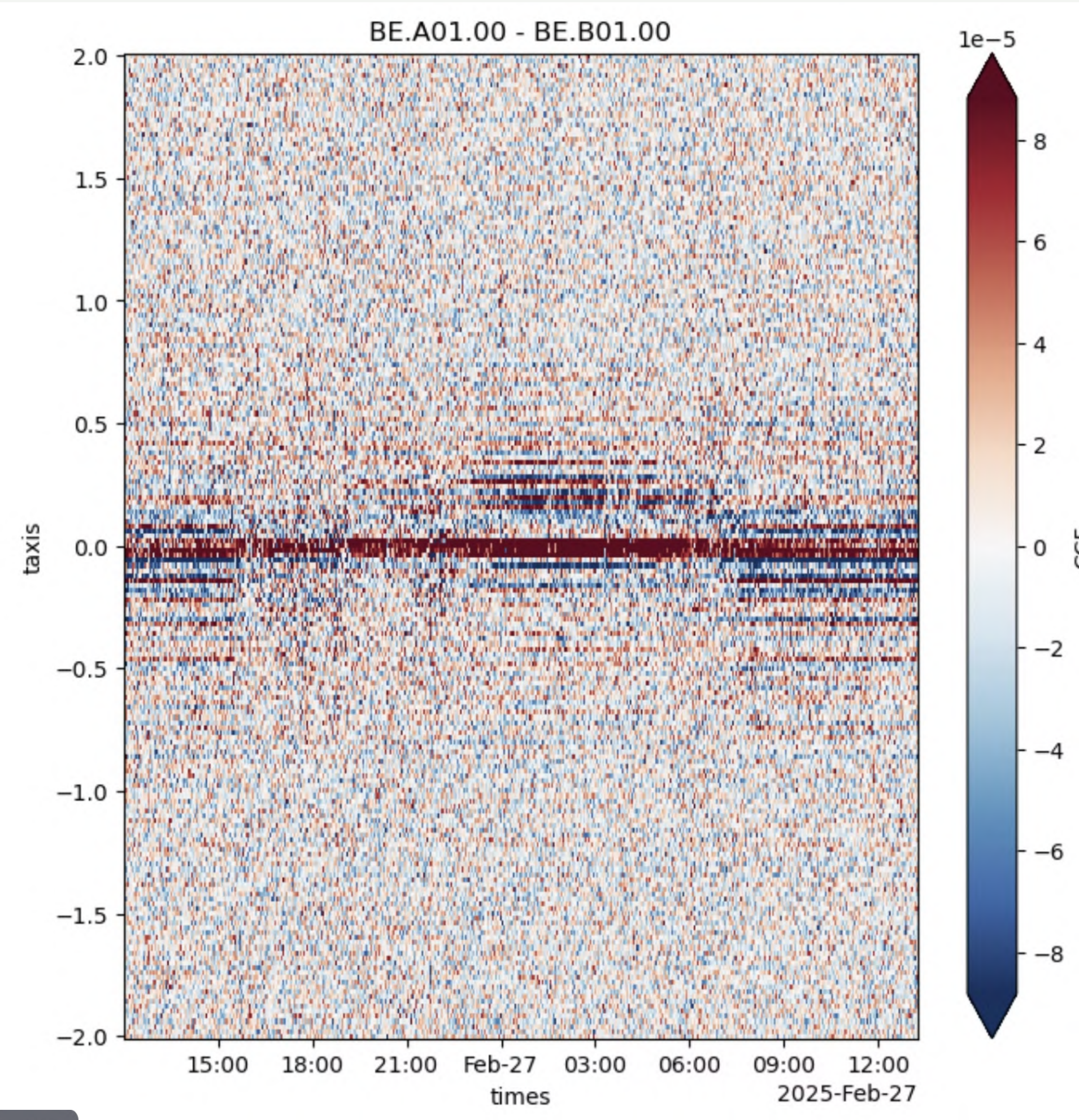
N Array

We lost
Heiner

but we
won the
skilift!

Free active
source





Next year ?



Koen

Next year ?

TODO EDIT



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